SHBT261 Mini Project 1

Cindy Sun

This code runs through OnDemand.

In [1]:
```
# Imports
import os, time, copy, json, warnings
import numpy as np
import pandas as pd
from pathlib import Path
from itertools import product

from PIL import Image
from skimage.feature import hog
from skimage.color import rgb2gray
from tqdm import tqdm

import matplotlib.pyplot as plt
import matplotlib.image as mpimg
import seaborn as sns

from sklearn.svm import SVC
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.metrics import (accuracy_score, confusion_matrix,
                             classification_report, top_k_accuracy_score)

import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import Dataset, DataLoader
from torchvision import models, transforms
import timm
import joblib
import shutil

warnings.filterwarnings('ignore')

print('All imports OK ✓')
print(f'PyTorch  : {torch.__version__}')
print(f'timm     : {timm.__version__}')
print(f'CUDA     : {torch.cuda.is_available()}')
print(f'MPS      : {torch.backends.mps.is_available()}')
```

```
All imports OK ✓
PyTorch  : 2.10.0
timm     : 1.0.25
CUDA     : True
MPS      : False
```

In [2]:
```
# Paths
print(os.getcwd())
```

```
print(os.path.expanduser("~"))

PROJECT_DIR = Path(os.path.expanduser("~")) / "SHBT261_miniproject1"
DATA_DIR    = PROJECT_DIR / "data" / "caltech-101"
FIG_DIR     = PROJECT_DIR / "figures" / "experiments"
MODEL_DIR   = PROJECT_DIR / "models" / "experiments"
RESULTS_DIR = PROJECT_DIR / "results"

for d in [FIG_DIR, MODEL_DIR, RESULTS_DIR]:
    d.mkdir(parents=True, exist_ok=True)

DEVICE = torch.device("mps"  if torch.backends.mps.is_available()
                 else "cuda" if torch.cuda.is_available()
                 else "cpu")

print(f"Project : {PROJECT_DIR}")
print(f"Data    : {DATA_DIR}")
print(f"Device  : {DEVICE}")
print(f"Data exists: {DATA_DIR.exists()}")
```

```
/shared/home/xis553/SHBT261_miniproject1
/shared/home/xis553
Project : /shared/home/xis553/SHBT261_miniproject1
Data    : /shared/home/xis553/SHBT261_miniproject1/data/caltech-101
Device  : cuda
Data exists: True
```

In [3]:

```
# Load Data & Split
records = []
for cls_folder in sorted(DATA_DIR.iterdir()):
    if cls_folder.is_dir() and not cls_folder.name.startswith('.'):
        for img_path in cls_folder.glob("*.jpg"):
            records.append({"image_path": str(img_path),
                            "class_name": cls_folder.name})

df = pd.DataFrame(records)
le = LabelEncoder()
df["label"] = le.fit_transform(df["class_name"])
NUM_CLASSES  = df["label"].nunique()

train_df, temp_df = train_test_split(df,      test_size=0.30, stratify=df["l
val_df,   test_df = train_test_split(temp_df, test_size=0.50, stratify=temp_
train_df = train_df.reset_index(drop=True)
val_df   = val_df.reset_index(drop=True)
test_df  = test_df.reset_index(drop=True)

print(f"Total images : {len(df)}")
print(f"Classes      : {NUM_CLASSES}")
print(f"Train        : {len(train_df)}")
print(f"Val          : {len(val_df)}")
print(f"Test         : {len(test_df)}")
```

```
Total images : 8677
Classes      : 101
Train        : 6073
Val          : 1302
Test         : 1302
```

In [4]:
```
# HOG Feature Extraction (SVM only)
HOG_CACHE = MODEL_DIR.parent.parent / "models"

def extract_hog(img_path, size):
    img      = Image.open(img_path).convert("RGB").resize(size)
    img_gray = rgb2gray(np.array(img))
    return hog(img_gray, orientations=9, pixels_per_cell=(8,8),
               cells_per_block=(2,2), feature_vector=True)

def build_hog_arrays(dataframe, size):
    X, y = [], []
    for _, row in tqdm(dataframe.iterrows(), total=len(dataframe), leave=Fal
        X.append(extract_hog(row["image_path"], size))
        y.append(row["label"])
    return np.array(X, dtype=np.float32), np.array(y)

def get_hog_arrays(size):
    """Load from cache or extract HOG features for a given image size."""
    tag = f"{size[0]}x{size[1]}"
    cache_files = {
        "X_train": MODEL_DIR / f"hog_{tag}_X_train.npy",
        "y_train": MODEL_DIR / f"hog_{tag}_y_train.npy",
        "X_val":   MODEL_DIR / f"hog_{tag}_X_val.npy",
        "y_val":   MODEL_DIR / f"hog_{tag}_y_val.npy",
        "X_test":  MODEL_DIR / f"hog_{tag}_X_test.npy",
        "y_test":  MODEL_DIR / f"hog_{tag}_y_test.npy",
    }
    if all(f.exists() for f in cache_files.values()):
        print(f"  HOG {tag} cache found — loading...")
        return {k: np.load(v) for k, v in cache_files.items()}

    print(f"  Extracting HOG {tag} features (first time)...")
    X_tr, y_tr = build_hog_arrays(train_df, size)
    X_va, y_va = build_hog_arrays(val_df,   size)
    X_te, y_te = build_hog_arrays(test_df,  size)

    arrays = {"X_train": X_tr, "y_train": y_tr,
              "X_val":   X_va, "y_val":   y_va,
              "X_test":  X_te, "y_test":  y_te}
    for k, v in arrays.items():
        np.save(cache_files[k], v)
    print(f"  HOG {tag} saved ✓")
    return arrays

print("HOG extraction functions ready ✓")
```

```
HOG extraction functions ready ✓
```

In [5]:
```
# Dataset & DataLoader Helpers (for DL models)
NUM_WORKERS = 12  # since i am using ondemand server
```

```
BATCH_SIZE  = 32

class Caltech101Dataset(Dataset):
    def __init__(self, dataframe, transform=None):
        self.df        = dataframe
        self.transform = transform

    def __len__(self):
        return len(self.df)

    def __getitem__(self, idx):
        row   = self.df.iloc[idx]
        image = Image.open(row["image_path"]).convert("RGB")
        if self.transform:
            image = self.transform(image)
        return image, torch.tensor(row["label"], dtype=torch.long)


def get_transforms(img_size, augment=True):
    """Returns train and val/test transforms."""
    mean = [0.485, 0.456, 0.406]
    std  = [0.229, 0.224, 0.225]

    if augment:
        train_tf = transforms.Compose([
            transforms.Resize((img_size, img_size)),
            transforms.RandomHorizontalFlip(),
            transforms.RandomRotation(10),
            transforms.ColorJitter(brightness=0.2, contrast=0.2, saturation=
            transforms.ToTensor(),
            transforms.Normalize(mean, std)
        ])
    else:
        train_tf = transforms.Compose([
            transforms.Resize((img_size, img_size)),
            transforms.ToTensor(),
            transforms.Normalize(mean, std)
        ])

    val_tf = transforms.Compose([
        transforms.Resize((img_size, img_size)),
        transforms.ToTensor(),
        transforms.Normalize(mean, std)
    ])
    return train_tf, val_tf


def get_loaders(img_size, augment=True):
    """Returns train, val, test DataLoaders."""
    train_tf, val_tf = get_transforms(img_size, augment)
    train_loader = DataLoader(Caltech101Dataset(train_df, train_tf),
                              batch_size=BATCH_SIZE, shuffle=True,
                              num_workers=NUM_WORKERS, pin_memory=True)
    val_loader   = DataLoader(Caltech101Dataset(val_df,   val_tf),
                              batch_size=BATCH_SIZE, shuffle=False,
                              num_workers=NUM_WORKERS, pin_memory=True)
```

```
    test_loader  = DataLoader(Caltech101Dataset(test_df,  val_tf),
                              batch_size=BATCH_SIZE, shuffle=False,
                              num_workers=NUM_WORKERS, pin_memory=True)
    return train_loader, val_loader, test_loader

print("Dataset helpers ready ✓")
```

```
Dataset helpers ready ✓
```

In [6]:

```
# Model Builders
def build_model(model_name, num_classes, freeze_backbone=True):
    """Build any supported model by name."""

    if model_name == "resnet18":
        model = models.resnet18(weights="IMAGENET1K_V1")
        if freeze_backbone:
            for p in model.parameters(): p.requires_grad = False
        model.fc = nn.Sequential(
            nn.Dropout(0.3),
            nn.Linear(model.fc.in_features, num_classes)
        )

    elif model_name == "efficientnet_b0":
        model = models.efficientnet_b0(weights="IMAGENET1K_V1")
        if freeze_backbone:
            for p in model.parameters(): p.requires_grad = False
        in_features = model.classifier[1].in_features
        model.classifier = nn.Sequential(
            nn.Dropout(0.3),
            nn.Linear(in_features, num_classes)
        )

    elif model_name == "vit_small":
        model = timm.create_model("vit_small_patch16_224",
                                  pretrained=True, num_classes=0)
        if freeze_backbone:
            for p in model.parameters(): p.requires_grad = False
        model.head = nn.Sequential(
            nn.LayerNorm(model.num_features),
            nn.Dropout(0.3),
            nn.Linear(model.num_features, num_classes)
        )

    else:
        raise ValueError(f"Unknown model: {model_name}")

    return model


def get_optimizer(opt_name, params, lr, weight_decay=1e-4):
    """Return optimizer by name."""
    if opt_name == "adam":
        return optim.Adam(params, lr=lr)
    elif opt_name == "adamw":
        return optim.AdamW(params, lr=lr, weight_decay=weight_decay)
    elif opt_name == "sgd":
```

```
        return optim.SGD(params, lr=lr, momentum=0.9,
                         weight_decay=weight_decay, nesterov=True)
    else:
        raise ValueError(f"Unknown optimizer: {opt_name}")

print("Model builders ready ✓")
```

```
Model builders ready ✓
```

In [7]:

```
# Training & Evaluation Functions
def train_one_epoch(model, loader, criterion, optimizer, device):
    model.train()
    total_loss, correct = 0.0, 0
    for images, labels in loader:
        images, labels = images.to(device), labels.long().to(device)
        optimizer.zero_grad()
        outputs = model(images)
        loss    = criterion(outputs, labels)
        loss.backward()
        optimizer.step()
        total_loss += loss.item() * images.size(0)
        correct    += (outputs.argmax(1) == labels).sum().item()
    n = len(loader.dataset)
    return total_loss / n, correct / n


@torch.no_grad()
def evaluate(model, loader, criterion, device):
    model.eval()
    total_loss, correct = 0.0, 0
    all_preds, all_labels, all_probs = [], [], []
    for images, labels in loader:
        images, labels = images.to(device), labels.long().to(device)
        outputs = model(images)
        loss    = criterion(outputs, labels)
        probs   = torch.softmax(outputs, dim=1)
        preds   = outputs.argmax(1)
        total_loss += loss.item() * images.size(0)
        correct    += (preds == labels).sum().item()
        all_preds.append(preds.cpu().numpy())
        all_labels.append(labels.cpu().numpy())
        all_probs.append(probs.cpu().numpy())
    n = len(loader.dataset)
    return (total_loss / n, correct / n,
            np.concatenate(all_preds),
            np.concatenate(all_labels),
            np.concatenate(all_probs))


def train_dl_model(model, train_loader, val_loader, cfg, device=DEVICE):
    """Full 2-phase training: freeze → train head, unfreeze → fine-tune."""
    history   = {"train_loss": [], "train_acc": [],
                 "val_loss":   [], "val_acc":   []}
    criterion = nn.CrossEntropyLoss(label_smoothing=0.1)
    best_acc, best_weights = 0.0, copy.deepcopy(model.state_dict())
```

```
    phases = [
        {"epochs": cfg["epochs"], "lr": cfg["lr"],      "unfreeze": False},
        {"epochs": cfg["epochs"], "lr": cfg["lr"] / 10, "unfreeze": True},
    ]

    for phase_idx, phase in enumerate(phases):
        if phase["unfreeze"]:
            for p in model.parameters():
                p.requires_grad = True

        trainable = filter(lambda p: p.requires_grad, model.parameters())
        optimizer = get_optimizer(cfg["optimizer"], trainable, phase["lr"])
        scheduler = optim.lr_scheduler.CosineAnnealingLR(
            optimizer, T_max=phase["epochs"]
        )

        desc = "Head only" if not phase["unfreeze"] else "Full fine-tune"
        print(f"  Phase {phase_idx+1} ({desc}) — {phase['epochs']} epochs")

        for epoch in range(phase["epochs"]):
            t0 = time.time()
            tr_loss, tr_acc = train_one_epoch(model, train_loader,
                                              criterion, optimizer, device)
            va_loss, va_acc, _, _, _ = evaluate(model, val_loader,
                                                criterion, device)
            scheduler.step()

            history["train_loss"].append(tr_loss)
            history["train_acc"].append(tr_acc)
            history["val_loss"].append(va_loss)
            history["val_acc"].append(va_acc)

            if va_acc > best_acc:
                best_acc     = va_acc
                best_weights = copy.deepcopy(model.state_dict())

            print(f"    Epoch [{epoch+1:02d}/{phase['epochs']}] "
                  f"Train {tr_acc*100:.1f}% | Val {va_acc*100:.1f}% "
                  f"| {time.time()-t0:.1f}s")

    model.load_state_dict(best_weights)
    return model, history, best_acc


def compute_metrics(y_true, y_pred, y_prob):
    """Compute all required metrics for the report."""
    acc   = accuracy_score(y_true, y_pred)
    top5  = top_k_accuracy_score(y_true, y_prob, k=5)
    cm    = confusion_matrix(y_true, y_pred)
    pca   = cm.diagonal() / cm.sum(axis=1)  # per-class accuracy
    report = classification_report(y_true, y_pred,
                                   target_names=le.classes_,
                                   digits=3, zero_division=0,
                                   output_dict=True)
    return {
        "accuracy":        acc,
```

```
        "top5_accuracy":   top5,
        "macro_f1":        report["macro avg"]["f1-score"],
        "weighted_f1":     report["weighted avg"]["f1-score"],
        "macro_precision": report["macro avg"]["precision"],
        "macro_recall":    report["macro avg"]["recall"],
        "per_class_acc":   pca.tolist(),
        "confusion_matrix": cm.tolist(),
    }

print("Training & evaluation functions ready ✓")
```

```
Training & evaluation functions ready ✓
```

In [8]:

```
# Experiment Configuration (manually input hyperparameters for gird search)

# SVM configurations (img_size only — other params fixed for practicality)
SVM_CONFIGS = [
    {"model": "svm", "img_size": 128, "C": 10,  "augment": False},
    {"model": "svm", "img_size": 128, "C": 10,  "augment": True},
    {"model": "svm", "img_size": 64,  "C": 10,  "augment": False},
    {"model": "svm", "img_size": 64,  "C": 10,  "augment": True}
]

# Deep Learning grid — all combinations will be generated
DL_MODELS     = ["resnet18", "efficientnet_b0", "vit_small"]
IMG_SIZES     = [128, 224]
EPOCHS_LIST   = [5, 10]
LR_LIST       = [1e-3, 1e-4]
OPTIMIZERS    = ["adam", "adamw", "sgd"]
AUGMENT_LIST  = [True, False]


# Generate all DL combinations
DL_CONFIGS = []
for model_name, img_size, epochs, lr, opt, aug in product(
        DL_MODELS, IMG_SIZES, EPOCHS_LIST, LR_LIST, OPTIMIZERS, AUGMENT_LIST

    # ViT requires 224 — skip 128 for ViT
    if model_name == "vit_small" and img_size == 128:
        continue

    DL_CONFIGS.append({
        "model":     model_name,
        "img_size":  img_size,
        "epochs":    epochs,
        "lr":        lr,
        "optimizer": opt,
        "augment":   aug,
    })

print(f"SVM experiments    : {len(SVM_CONFIGS)}")
print(f"DL experiments     : {len(DL_CONFIGS)}")
print(f"Total experiments  : {len(SVM_CONFIGS) + len(DL_CONFIGS)}")
print()
print("Sample DL configs:")
```

```
for c in DL_CONFIGS[:5]:
    print(f"  {c}")
```

```
SVM experiments    : 4
DL experiments     : 120
Total experiments  : 124

Sample DL configs:
  {'model': 'resnet18', 'img_size': 128, 'epochs': 5, 'lr': 0.001, 'optimize
r': 'adam', 'augment': True}
  {'model': 'resnet18', 'img_size': 128, 'epochs': 5, 'lr': 0.001, 'optimize
r': 'adam', 'augment': False}
  {'model': 'resnet18', 'img_size': 128, 'epochs': 5, 'lr': 0.001, 'optimize
r': 'adamw', 'augment': True}
  {'model': 'resnet18', 'img_size': 128, 'epochs': 5, 'lr': 0.001, 'optimize
r': 'adamw', 'augment': False}
  {'model': 'resnet18', 'img_size': 128, 'epochs': 5, 'lr': 0.001, 'optimize
r': 'sgd', 'augment': True}
```

In [9]:

```
# Results Tracker
RESULTS_FILE = RESULTS_DIR / "all_results.json"

def load_results():
    if RESULTS_FILE.exists():
        with open(RESULTS_FILE) as f:
            return json.load(f)
    return {}

def save_results(results):
    with open(RESULTS_FILE, "w") as f:
        json.dump(results, f, indent=2)

def make_run_key(cfg):
    """Unique string key for each config."""
    if cfg["model"] == "svm":
        return (f"svm_img{cfg['img_size']}_C{cfg['C']}"
                f"_aug{cfg['augment']}")
    return (f"{cfg['model']}_img{cfg['img_size']}"
            f"_ep{cfg['epochs']}_lr{cfg['lr']}"
            f"_{cfg['optimizer']}_aug{cfg['augment']}")

# Load any existing results (resumes from checkpoint)
all_results = load_results()
print(f"Existing completed runs: {len(all_results)}")
```

```
Existing completed runs: 124
```

In [10]:

```
# Run SVM Experiments
print("="*60)
print("RUNNING SVM EXPERIMENTS")
print("="*60)

for cfg in SVM_CONFIGS:
    run_key = make_run_key(cfg)

    if run_key in all_results:
        print(f"  SKIP (already done): {run_key}")
```

```
        continue

    print(f"\nRunning: {run_key}")
    t_start = time.time()

    # Get HOG features
    size    = (cfg["img_size"], cfg["img_size"])
    arrays  = get_hog_arrays(size)

    X_tr = arrays["X_train"].copy()
    y_tr = arrays["y_train"]
    X_te = arrays["X_test"].copy()
    y_te = arrays["y_test"]

    # Augment: add flipped versions to train
    if cfg["augment"]:
        print("  Extracting augmented HOG features...")
        X_aug, y_aug = [], []
        for _, row in tqdm(train_df.iterrows(), total=len(train_df), leave=F
            img  = Image.open(row["image_path"]).convert("RGB").resize(size)
            flip = img.transpose(Image.FLIP_LEFT_RIGHT)
            for im in [img, flip]:
                gray = rgb2gray(np.array(im))
                X_aug.append(hog(gray, orientations=9, pixels_per_cell=(8,8)
                                 cells_per_block=(2,2), feature_vector=True)
                y_aug.append(row["label"])
        X_tr = np.array(X_aug, dtype=np.float32)
        y_tr = np.array(y_aug)

    # Scale
    scaler = StandardScaler()
    X_tr_sc = scaler.fit_transform(X_tr)
    X_te_sc = scaler.transform(X_te)

    # Train SVM
    print("  Training SVM (this takes a while)...")
    svm = SVC(kernel="rbf", C=cfg["C"], gamma="scale",
              probability=True, random_state=42)
    svm.fit(X_tr_sc, y_tr)

    # Evaluate
    y_pred = svm.predict(X_te_sc)
    y_prob = svm.predict_proba(X_te_sc)
    metrics = compute_metrics(y_te, y_pred, y_prob)

    # Save model
    joblib.dump(svm,    MODEL_DIR / f"{run_key}_svm.pkl")
    joblib.dump(scaler, MODEL_DIR / f"{run_key}_scaler.pkl")

    # Record results
    elapsed = time.time() - t_start
    all_results[run_key] = {
        "config":       cfg,
        "metrics":      {k: v for k, v in metrics.items()
                         if k not in ["per_class_acc", "confusion_matrix"]},
        "per_class_acc": metrics["per_class_acc"],
```

```
        "confusion_matrix": metrics["confusion_matrix"],
        "train_time_sec": elapsed,
    }
    save_results(all_results)
    print(f"  ✓ Done — Acc: {metrics['accuracy']*100:.2f}% "
          f"| Top-5: {metrics['top5_accuracy']*100:.2f}% "
          f"| Time: {elapsed/60:.1f} min")

print("\nAll SVM experiments complete ✓")
```

```
============================================================
RUNNING SVM EXPERIMENTS
============================================================
  SKIP (already done): svm_img128_C10_augFalse
  SKIP (already done): svm_img128_C10_augTrue
  SKIP (already done): svm_img64_C10_augFalse
  SKIP (already done): svm_img64_C10_augTrue

All SVM experiments complete ✓
```

In [11]:

```
# Run DL Experiments
print("="*60)
print("RUNNING DEEP LEARNING EXPERIMENTS")
print(f"Total configs: {len(DL_CONFIGS)}")
print("="*60)

criterion = nn.CrossEntropyLoss(label_smoothing=0.1)

for idx, cfg in enumerate(DL_CONFIGS):
    run_key = make_run_key(cfg)

    if run_key in all_results:
        print(f"[{idx+1}/{len(DL_CONFIGS)}] SKIP: {run_key}")
        continue

    print(f"\n[{idx+1}/{len(DL_CONFIGS)}] Running: {run_key}")
    t_start = time.time()

    try:
        # Build loaders
        train_loader, val_loader, test_loader = get_loaders(
            cfg["img_size"], cfg["augment"]
        )

        # Build model
        model = build_model(cfg["model"], NUM_CLASSES,
                            freeze_backbone=True).to(DEVICE)

        # Train
        model, history, best_val_acc = train_dl_model(
            model, train_loader, val_loader, cfg, DEVICE
        )

        # Evaluate on test set
        _, _, y_pred, y_true, y_prob = evaluate(
            model, test_loader, criterion, DEVICE
```

```
        )
        metrics = compute_metrics(y_true, y_pred, y_prob)

        # Save model weights
        torch.save(model.state_dict(), MODEL_DIR / f"{run_key}.pth")

        # Save training curves figure
        epochs_range = range(1, len(history["train_acc"]) + 1)
        fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 4))
        ax1.plot(epochs_range, [a*100 for a in history["train_acc"]], label=
        ax1.plot(epochs_range, [a*100 for a in history["val_acc"]],   label=
        ax1.axvline(x=cfg["epochs"], color="gray", linestyle="--", label="Ph
        ax1.set_title(f"{run_key}\nAccuracy"); ax1.set_ylabel("Acc (%)"); ax
        ax2.plot(epochs_range, history["train_loss"], label="Train")
        ax2.plot(epochs_range, history["val_loss"],   label="Val")
        ax2.axvline(x=cfg["epochs"], color="gray", linestyle="--")
        ax2.set_title("Loss"); ax2.set_ylabel("Loss"); ax2.legend()
        plt.tight_layout()
        plt.savefig(FIG_DIR / f"{run_key}_curves.png", dpi=100)
        plt.close()

        # Record results
        elapsed = time.time() - t_start
        all_results[run_key] = {
            "config":       cfg,
            "metrics":      {k: v for k, v in metrics.items()
                             if k not in ["per_class_acc", "confusion_matrix
            "per_class_acc":    metrics["per_class_acc"],
            "confusion_matrix": metrics["confusion_matrix"],
            "best_val_acc":  best_val_acc,
            "history":       history,
            "train_time_sec": elapsed,
        }
        save_results(all_results)
        print(f"  ✓ Acc: {metrics['accuracy']*100:.2f}% "
              f"| Top-5: {metrics['top5_accuracy']*100:.2f}% "
              f"| F1: {metrics['macro_f1']:.3f} "
              f"| Time: {elapsed/60:.1f} min")

        # Free GPU memory
        del model
        if torch.cuda.is_available(): torch.cuda.empty_cache()

    except Exception as e:
        print(f"  ✗ FAILED: {e}")
        all_results[run_key] = {"config": cfg, "error": str(e)}
        save_results(all_results)
        continue

print("\nAll DL experiments complete ✓")
```

```
============================================================
RUNNING DEEP LEARNING EXPERIMENTS
Total configs: 120
============================================================
[1/120] SKIP: resnet18_img128_ep5_lr0.001_adam_augTrue
[2/120] SKIP: resnet18_img128_ep5_lr0.001_adam_augFalse
[3/120] SKIP: resnet18_img128_ep5_lr0.001_adamw_augTrue
[4/120] SKIP: resnet18_img128_ep5_lr0.001_adamw_augFalse
[5/120] SKIP: resnet18_img128_ep5_lr0.001_sgd_augTrue
[6/120] SKIP: resnet18_img128_ep5_lr0.001_sgd_augFalse
[7/120] SKIP: resnet18_img128_ep5_lr0.0001_adam_augTrue
[8/120] SKIP: resnet18_img128_ep5_lr0.0001_adam_augFalse
[9/120] SKIP: resnet18_img128_ep5_lr0.0001_adamw_augTrue
[10/120] SKIP: resnet18_img128_ep5_lr0.0001_adamw_augFalse
[11/120] SKIP: resnet18_img128_ep5_lr0.0001_sgd_augTrue
[12/120] SKIP: resnet18_img128_ep5_lr0.0001_sgd_augFalse
[13/120] SKIP: resnet18_img128_ep10_lr0.001_adam_augTrue
[14/120] SKIP: resnet18_img128_ep10_lr0.001_adam_augFalse
[15/120] SKIP: resnet18_img128_ep10_lr0.001_adamw_augTrue
[16/120] SKIP: resnet18_img128_ep10_lr0.001_adamw_augFalse
[17/120] SKIP: resnet18_img128_ep10_lr0.001_sgd_augTrue
[18/120] SKIP: resnet18_img128_ep10_lr0.001_sgd_augFalse
[19/120] SKIP: resnet18_img128_ep10_lr0.0001_adam_augTrue
[20/120] SKIP: resnet18_img128_ep10_lr0.0001_adam_augFalse
[21/120] SKIP: resnet18_img128_ep10_lr0.0001_adamw_augTrue
[22/120] SKIP: resnet18_img128_ep10_lr0.0001_adamw_augFalse
[23/120] SKIP: resnet18_img128_ep10_lr0.0001_sgd_augTrue
[24/120] SKIP: resnet18_img128_ep10_lr0.0001_sgd_augFalse
[25/120] SKIP: resnet18_img224_ep5_lr0.001_adam_augTrue
[26/120] SKIP: resnet18_img224_ep5_lr0.001_adam_augFalse
[27/120] SKIP: resnet18_img224_ep5_lr0.001_adamw_augTrue
[28/120] SKIP: resnet18_img224_ep5_lr0.001_adamw_augFalse
[29/120] SKIP: resnet18_img224_ep5_lr0.001_sgd_augTrue
[30/120] SKIP: resnet18_img224_ep5_lr0.001_sgd_augFalse
[31/120] SKIP: resnet18_img224_ep5_lr0.0001_adam_augTrue
[32/120] SKIP: resnet18_img224_ep5_lr0.0001_adam_augFalse
[33/120] SKIP: resnet18_img224_ep5_lr0.0001_adamw_augTrue
[34/120] SKIP: resnet18_img224_ep5_lr0.0001_adamw_augFalse
[35/120] SKIP: resnet18_img224_ep5_lr0.0001_sgd_augTrue
[36/120] SKIP: resnet18_img224_ep5_lr0.0001_sgd_augFalse
[37/120] SKIP: resnet18_img224_ep10_lr0.001_adam_augTrue
[38/120] SKIP: resnet18_img224_ep10_lr0.001_adam_augFalse
[39/120] SKIP: resnet18_img224_ep10_lr0.001_adamw_augTrue
[40/120] SKIP: resnet18_img224_ep10_lr0.001_adamw_augFalse
[41/120] SKIP: resnet18_img224_ep10_lr0.001_sgd_augTrue
[42/120] SKIP: resnet18_img224_ep10_lr0.001_sgd_augFalse
[43/120] SKIP: resnet18_img224_ep10_lr0.0001_adam_augTrue
[44/120] SKIP: resnet18_img224_ep10_lr0.0001_adam_augFalse
[45/120] SKIP: resnet18_img224_ep10_lr0.0001_adamw_augTrue
[46/120] SKIP: resnet18_img224_ep10_lr0.0001_adamw_augFalse
[47/120] SKIP: resnet18_img224_ep10_lr0.0001_sgd_augTrue
[48/120] SKIP: resnet18_img224_ep10_lr0.0001_sgd_augFalse
[49/120] SKIP: efficientnet_b0_img128_ep5_lr0.001_adam_augTrue
[50/120] SKIP: efficientnet_b0_img128_ep5_lr0.001_adam_augFalse
[51/120] SKIP: efficientnet_b0_img128_ep5_lr0.001_adamw_augTrue
[52/120] SKIP: efficientnet_b0_img128_ep5_lr0.001_adamw_augFalse
```

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[53/120] SKIP: efficientnet_b0_img128_ep5_lr0.001_sgd_augTrue
[54/120] SKIP: efficientnet_b0_img128_ep5_lr0.001_sgd_augFalse
[55/120] SKIP: efficientnet_b0_img128_ep5_lr0.0001_adam_augTrue
[56/120] SKIP: efficientnet_b0_img128_ep5_lr0.0001_adam_augFalse
[57/120] SKIP: efficientnet_b0_img128_ep5_lr0.0001_adamw_augTrue
[58/120] SKIP: efficientnet_b0_img128_ep5_lr0.0001_adamw_augFalse
[59/120] SKIP: efficientnet_b0_img128_ep5_lr0.0001_sgd_augTrue
[60/120] SKIP: efficientnet_b0_img128_ep5_lr0.0001_sgd_augFalse
[61/120] SKIP: efficientnet_b0_img128_ep10_lr0.001_adam_augTrue
[62/120] SKIP: efficientnet_b0_img128_ep10_lr0.001_adam_augFalse
[63/120] SKIP: efficientnet_b0_img128_ep10_lr0.001_adamw_augTrue
[64/120] SKIP: efficientnet_b0_img128_ep10_lr0.001_adamw_augFalse
[65/120] SKIP: efficientnet_b0_img128_ep10_lr0.001_sgd_augTrue
[66/120] SKIP: efficientnet_b0_img128_ep10_lr0.001_sgd_augFalse
[67/120] SKIP: efficientnet_b0_img128_ep10_lr0.0001_adam_augTrue
[68/120] SKIP: efficientnet_b0_img128_ep10_lr0.0001_adam_augFalse
[69/120] SKIP: efficientnet_b0_img128_ep10_lr0.0001_adamw_augTrue
[70/120] SKIP: efficientnet_b0_img128_ep10_lr0.0001_adamw_augFalse
[71/120] SKIP: efficientnet_b0_img128_ep10_lr0.0001_sgd_augTrue
[72/120] SKIP: efficientnet_b0_img128_ep10_lr0.0001_sgd_augFalse
[73/120] SKIP: efficientnet_b0_img224_ep5_lr0.001_adam_augTrue
[74/120] SKIP: efficientnet_b0_img224_ep5_lr0.001_adam_augFalse
[75/120] SKIP: efficientnet_b0_img224_ep5_lr0.001_adamw_augTrue
[76/120] SKIP: efficientnet_b0_img224_ep5_lr0.001_adamw_augFalse
[77/120] SKIP: efficientnet_b0_img224_ep5_lr0.001_sgd_augTrue
[78/120] SKIP: efficientnet_b0_img224_ep5_lr0.001_sgd_augFalse
[79/120] SKIP: efficientnet_b0_img224_ep5_lr0.0001_adam_augTrue
[80/120] SKIP: efficientnet_b0_img224_ep5_lr0.0001_adam_augFalse
[81/120] SKIP: efficientnet_b0_img224_ep5_lr0.0001_adamw_augTrue
[82/120] SKIP: efficientnet_b0_img224_ep5_lr0.0001_adamw_augFalse
[83/120] SKIP: efficientnet_b0_img224_ep5_lr0.0001_sgd_augTrue
[84/120] SKIP: efficientnet_b0_img224_ep5_lr0.0001_sgd_augFalse
[85/120] SKIP: efficientnet_b0_img224_ep10_lr0.001_adam_augTrue
[86/120] SKIP: efficientnet_b0_img224_ep10_lr0.001_adam_augFalse
[87/120] SKIP: efficientnet_b0_img224_ep10_lr0.001_adamw_augTrue
[88/120] SKIP: efficientnet_b0_img224_ep10_lr0.001_adamw_augFalse
[89/120] SKIP: efficientnet_b0_img224_ep10_lr0.001_sgd_augTrue
[90/120] SKIP: efficientnet_b0_img224_ep10_lr0.001_sgd_augFalse
[91/120] SKIP: efficientnet_b0_img224_ep10_lr0.0001_adam_augTrue
[92/120] SKIP: efficientnet_b0_img224_ep10_lr0.0001_adam_augFalse
[93/120] SKIP: efficientnet_b0_img224_ep10_lr0.0001_adamw_augTrue
[94/120] SKIP: efficientnet_b0_img224_ep10_lr0.0001_adamw_augFalse
[95/120] SKIP: efficientnet_b0_img224_ep10_lr0.0001_sgd_augTrue
[96/120] SKIP: efficientnet_b0_img224_ep10_lr0.0001_sgd_augFalse
[97/120] SKIP: vit_small_img224_ep5_lr0.001_adam_augTrue
[98/120] SKIP: vit_small_img224_ep5_lr0.001_adam_augFalse
[99/120] SKIP: vit_small_img224_ep5_lr0.001_adamw_augTrue
[100/120] SKIP: vit_small_img224_ep5_lr0.001_adamw_augFalse
[101/120] SKIP: vit_small_img224_ep5_lr0.001_sgd_augTrue
[102/120] SKIP: vit_small_img224_ep5_lr0.001_sgd_augFalse
[103/120] SKIP: vit_small_img224_ep5_lr0.0001_adam_augTrue
[104/120] SKIP: vit_small_img224_ep5_lr0.0001_adam_augFalse
[105/120] SKIP: vit_small_img224_ep5_lr0.0001_adamw_augTrue
[106/120] SKIP: vit_small_img224_ep5_lr0.0001_adamw_augFalse
[107/120] SKIP: vit_small_img224_ep5_lr0.0001_sgd_augTrue
[108/120] SKIP: vit_small_img224_ep5_lr0.0001_sgd_augFalse
```

```
[109/120] SKIP: vit_small_img224_ep10_lr0.001_adam_augTrue
[110/120] SKIP: vit_small_img224_ep10_lr0.001_adam_augFalse
[111/120] SKIP: vit_small_img224_ep10_lr0.001_adamw_augTrue
[112/120] SKIP: vit_small_img224_ep10_lr0.001_adamw_augFalse
[113/120] SKIP: vit_small_img224_ep10_lr0.001_sgd_augTrue
[114/120] SKIP: vit_small_img224_ep10_lr0.001_sgd_augFalse
[115/120] SKIP: vit_small_img224_ep10_lr0.0001_adam_augTrue
[116/120] SKIP: vit_small_img224_ep10_lr0.0001_adam_augFalse
[117/120] SKIP: vit_small_img224_ep10_lr0.0001_adamw_augTrue
[118/120] SKIP: vit_small_img224_ep10_lr0.0001_adamw_augFalse
[119/120] SKIP: vit_small_img224_ep10_lr0.0001_sgd_augTrue
[120/120] SKIP: vit_small_img224_ep10_lr0.0001_sgd_augFalse

All DL experiments complete ✓
```

In [12]:

```python
# Results Summary Table
all_results = load_results()  # reload in case re-run this cell

rows = []
for key, result in all_results.items():
    if "error" in result:
        continue
    cfg = result["config"]
    m   = result["metrics"]
    rows.append({
        "run_key":        key,
        "model":          cfg["model"],
        "img_size":       cfg["img_size"],
        "epochs":         cfg.get("epochs", "N/A"),
        "lr":             cfg.get("lr", "N/A"),
        "optimizer":      cfg.get("optimizer", "rbf"),
        "augment":        cfg["augment"],
        "test_acc_%":     round(m["accuracy"]        * 100, 2),
        "top5_acc_%":     round(m["top5_accuracy"]   * 100, 2),
        "macro_f1":       round(m["macro_f1"],            3),
        "weighted_f1":    round(m["weighted_f1"],         3),
        "macro_precision":round(m["macro_precision"],     3),
        "macro_recall":   round(m["macro_recall"],        3),
        "train_min":      round(result["train_time_sec"] / 60, 1),
    })

results_df = pd.DataFrame(rows).sort_values("test_acc_%", ascending=False)
results_df.to_csv(RESULTS_DIR / "summary_table.csv", index=False)

print(f"Total completed runs: {len(results_df)}")
print("\nTop 10 configurations by test accuracy:")
display(results_df.head(10))
```

```
Total completed runs: 124

Top 10 configurations by test accuracy:
```

	run_key	model	img_size	epochs	lr
114	vit_small_img224_ep10_lr0.001_adamw_augFalse	vit_small	224	10	0.001
113	vit_small_img224_ep10_lr0.001_adamw_augTrue	vit_small	224	10	0.001
111	vit_small_img224_ep10_lr0.001_adam_augTrue	vit_small	224	10	0.001
112	vit_small_img224_ep10_lr0.001_adam_augFalse	vit_small	224	10	0.001
117	vit_small_img224_ep10_lr0.0001_adam_augTrue	vit_small	224	10	0.0001
101	vit_small_img224_ep5_lr0.001_adamw_augTrue	vit_small	224	5	0.001
102	vit_small_img224_ep5_lr0.001_adamw_augFalse	vit_small	224	5	0.001
119	vit_small_img224_ep10_lr0.0001_adamw_augTrue	vit_small	224	10	0.0001
120	vit_small_img224_ep10_lr0.0001_adamw_augFalse	vit_small	224	10	0.0001
99	vit_small_img224_ep5_lr0.001_adam_augTrue	vit_small	224	5	0.001

In [13]:
```
# Save Best Model per Architecture
BEST_MODELS_DIR = MODEL_DIR.parent / "best_models"
BEST_MODELS_DIR.mkdir(parents=True, exist_ok=True)

for model_name in ["svm", "resnet18", "efficientnet_b0", "vit_small"]:
    subset = results_df[results_df["model"] == model_name]
    if subset.empty:
        print(f"  {model_name}: no results yet — skipping")
        continue

    best_key = subset.iloc[0]["run_key"]   # already sorted by accuracy

    if model_name == "svm":
        # SVM saves as .pkl
        src_model  = MODEL_DIR / f"{best_key}_svm.pkl"
        src_scaler = MODEL_DIR / f"{best_key}_scaler.pkl"
        dst_model  = BEST_MODELS_DIR / f"best_svm.pkl"
        dst_scaler = BEST_MODELS_DIR / f"best_svm_scaler.pkl"
        if src_model.exists():
            shutil.copy2(src_model,  dst_model)
            shutil.copy2(src_scaler, dst_scaler)
    else:
        # DL models save as .pth
        src = MODEL_DIR / f"{best_key}.pth"
        dst = BEST_MODELS_DIR / f"best_{model_name}.pth"
        if src.exists():
            shutil.copy2(src, dst)

    acc = subset.iloc[0]["test_acc_%"]
    print(f"  ✓ {model_name}: {best_key}")
    print(f"      Accuracy: {acc}% → saved to best_models/")

# Save a README so I know which config each best model came from
readme_lines = ["# Best Models\n\n"]
for model_name in ["svm", "resnet18", "efficientnet_b0", "vit_small"]:
```

```
    subset = results_df[results_df["model"] == model_name]
    if subset.empty:
        continue
    best = subset.iloc[0]
    readme_lines.append(f"## {model_name}\n")
    readme_lines.append(f"- run_key  : {best['run_key']}\n")
    readme_lines.append(f"- accuracy : {best['test_acc_%']}%\n")
    readme_lines.append(f"- top5_acc : {best['top5_acc_%']}%\n")
    readme_lines.append(f"- macro_f1 : {best['macro_f1']}\n")
    readme_lines.append(f"- config   : img={best['img_size']}, "
                        f"epochs={best['epochs']}, lr={best['lr']}, "
                        f"opt={best['optimizer']}, aug={best['augment']}\n\r

with open(BEST_MODELS_DIR / "README.md", "w") as f:
    f.writelines(readme_lines)

print(f"\nBest models saved to: {BEST_MODELS_DIR}")
print("README.md written ✓")
```

```
 ✓ svm: svm_img64_C10_augFalse
     Accuracy: 67.36% → saved to best_models/
 ✓ resnet18: resnet18_img224_ep5_lr0.001_adamw_augFalse
     Accuracy: 97.0% → saved to best_models/
 ✓ efficientnet_b0: efficientnet_b0_img224_ep10_lr0.001_adam_augTrue
     Accuracy: 97.39% → saved to best_models/
 ✓ vit_small: vit_small_img224_ep10_lr0.001_adamw_augFalse
     Accuracy: 98.23% → saved to best_models/

Best models saved to: /shared/home/xis553/SHBT261_miniproject1/models/best_m
odels
README.md written ✓
```

In [14]:

```
# Best Model Per Architecture
print("Best configuration per model:\n")
for model_name in ["svm", "resnet18", "efficientnet_b0", "vit_small"]:
    subset = results_df[results_df["model"] == model_name]
    if subset.empty:
        print(f"  {model_name}: no results yet")
        continue
    best = subset.iloc[0]
    print(f"  {model_name}:")
    print(f"    Accuracy  : {best['test_acc_%']}%")
    print(f"    Top-5     : {best['top5_acc_%']}%")
    print(f"    Macro F1  : {best['macro_f1']}")
    print(f"    Config    : img={best['img_size']}, "
          f"epochs={best['epochs']}, lr={best['lr']}, "
          f"opt={best['optimizer']}, aug={best['augment']}")
    print(f"    Train time: {best['train_min']} min\n")
```

```
Best configuration per model:

  svm:
    Accuracy  : 67.36%
    Top-5     : 84.18%
    Macro F1  : 0.531
    Config    : img=64, epochs=N/A, lr=N/A, opt=rbf, aug=False
    Train time: 3.8 min

  resnet18:
    Accuracy  : 97.0%
    Top-5     : 99.77%
    Macro F1  : 0.954
    Config    : img=224, epochs=5, lr=0.001, opt=adamw, aug=False
    Train time: 1.3 min

  efficientnet_b0:
    Accuracy  : 97.39%
    Top-5     : 99.85%
    Macro F1  : 0.962
    Config    : img=224, epochs=10, lr=0.001, opt=adam, aug=True
    Train time: 4.5 min

  vit_small:
    Accuracy  : 98.23%
    Top-5     : 99.69%
    Macro F1  : 0.974
    Config    : img=224, epochs=10, lr=0.001, opt=adamw, aug=False
    Train time: 7.5 min
```

In [15]:

```
# Comparison Bar Chart

MODEL_ORDER = ["svm", "resnet18", "efficientnet_b0", "vit_small"]
MODEL_LABELS = {"svm": "SVM", "resnet18": "ResNet-18",
                "efficientnet_b0": "EfficientNet-B0", "vit_small": "ViT-Smal

# Best accuracy per model — reordered
best_per_model = results_df.groupby("model")["test_acc_%"].max().reset_index
best_per_model["model"] = pd.Categorical(best_per_model["model"], categories
best_per_model = best_per_model.sort_values("model")

colors = {"svm": "#5bc0de", "resnet18": "#5cb85c",
          "efficientnet_b0": "#f0ad4e", "vit_small": "#9b59b6"}

TITLE_SIZE  = 16
LABEL_SIZE  = 14
TICK_SIZE   = 13
ANNOT_SIZE  = 13
LEGEND_SIZE = 12

fig, axes = plt.subplots(1, 3, figsize=(22, 6))

# ── Plot 1: Best accuracy per model ──────────────────────────
ax = axes[0]
bars = ax.bar(
```

```
    [MODEL_LABELS[m] for m in best_per_model["model"]],
    best_per_model["test_acc_%"],
    color=[colors.get(m, "gray") for m in best_per_model["model"]]
)
for bar, val in zip(bars, best_per_model["test_acc_%"]):
    ax.text(bar.get_x() + bar.get_width() / 2, val + 0.3,
            f"{val:.1f}%", ha="center", fontsize=ANNOT_SIZE, fontweight="bol
ax.set_title("Best Test Accuracy per Model", fontsize=TITLE_SIZE, fontweight
ax.set_ylabel("Test Accuracy (%)", fontsize=LABEL_SIZE)
ax.set_xlabel("Model", fontsize=LABEL_SIZE)
ax.set_ylim(0, 105)
ax.tick_params(axis="both", labelsize=TICK_SIZE)

# ── Plot 2: Accuracy vs Training Time (scatter) ──────────────────
ax = axes[1]
for model_name in MODEL_ORDER:
    sub = results_df[results_df["model"] == model_name]
    ax.scatter(sub["train_min"], sub["test_acc_%"],
               label=MODEL_LABELS[model_name], alpha=0.6,
               color=colors.get(model_name, "gray"), s=60)
ax.set_xlabel("Training Time (min)", fontsize=LABEL_SIZE)
ax.set_ylabel("Test Accuracy (%)", fontsize=LABEL_SIZE)
ax.set_title("Accuracy vs Training Time", fontsize=TITLE_SIZE, fontweight="b
ax.tick_params(axis="both", labelsize=TICK_SIZE)
ax.legend(fontsize=LEGEND_SIZE)

# ── Plot 3: Augmentation effect ──────────────────────────────────
ax = axes[2]
aug_effect = results_df.groupby(["model", "augment"])["test_acc_%"].mean().u
aug_effect.index = pd.CategoricalIndex(aug_effect.index, categories=MODEL_OR
aug_effect = aug_effect.sort_index()

if True in aug_effect.columns and False in aug_effect.columns:
    x = np.arange(len(aug_effect))
    w = 0.35

    bars_no_aug   = ax.bar(x - w/2, aug_effect[False], width=w, label="No Au
    bars_with_aug = ax.bar(x + w/2, aug_effect[True],  width=w, label="With

    for bar, val in zip(bars_no_aug, aug_effect[False]):
        ax.text(bar.get_x() + bar.get_width() / 2, val + 0.3,
                f"{val:.1f}%", ha="center", va="bottom", fontsize=9, fontwei

    for bar, val in zip(bars_with_aug, aug_effect[True]):
        ax.text(bar.get_x() + bar.get_width() / 2, val + 0.3,
                f"{val:.1f}%", ha="center", va="bottom", fontsize=9, fontwei

    ax.set_xticks(x)
    ax.set_xticklabels([MODEL_LABELS[m] for m in aug_effect.index],
                       rotation=15, ha="right", fontsize=TICK_SIZE)
    ax.set_ylabel("Avg Test Accuracy (%)", fontsize=LABEL_SIZE)
    ax.set_xlabel("Model", fontsize=LABEL_SIZE)
    ax.set_title("Augmentation Effect per Model", fontsize=TITLE_SIZE, fontw
    ax.tick_params(axis="y", labelsize=TICK_SIZE)
    ax.legend(fontsize=LEGEND_SIZE)
    ax.set_ylim(0, aug_effect.values.max() + 8)  # extra headroom for annota
```

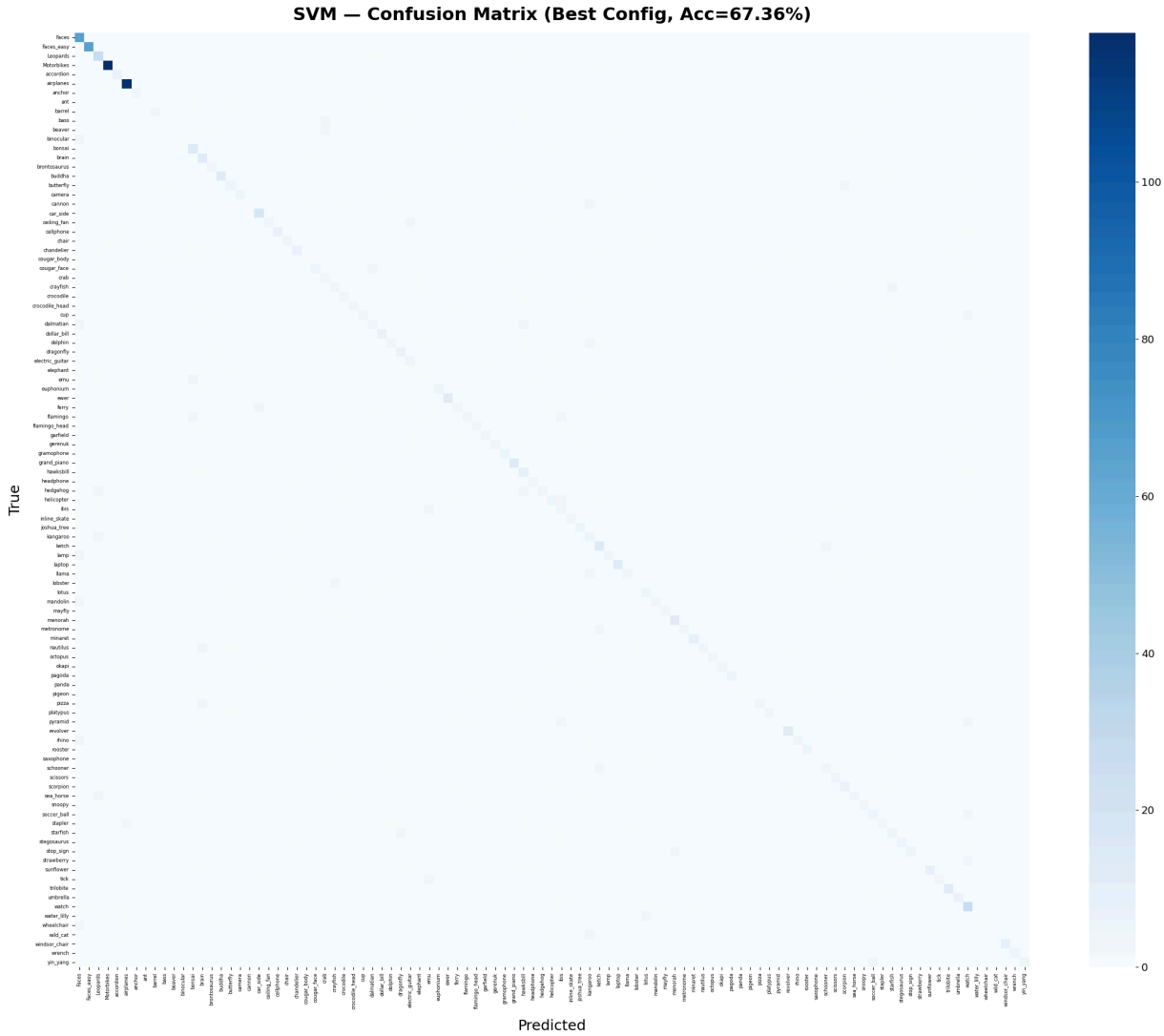
```
plt.tight_layout()
plt.savefig(FIG_DIR / "comparison_summary.png", dpi=150, bbox_inches="tight"
plt.show()
print("Saved: comparison_summary.png ✓")
```

```
Saved: comparison_summary.png ✓
```

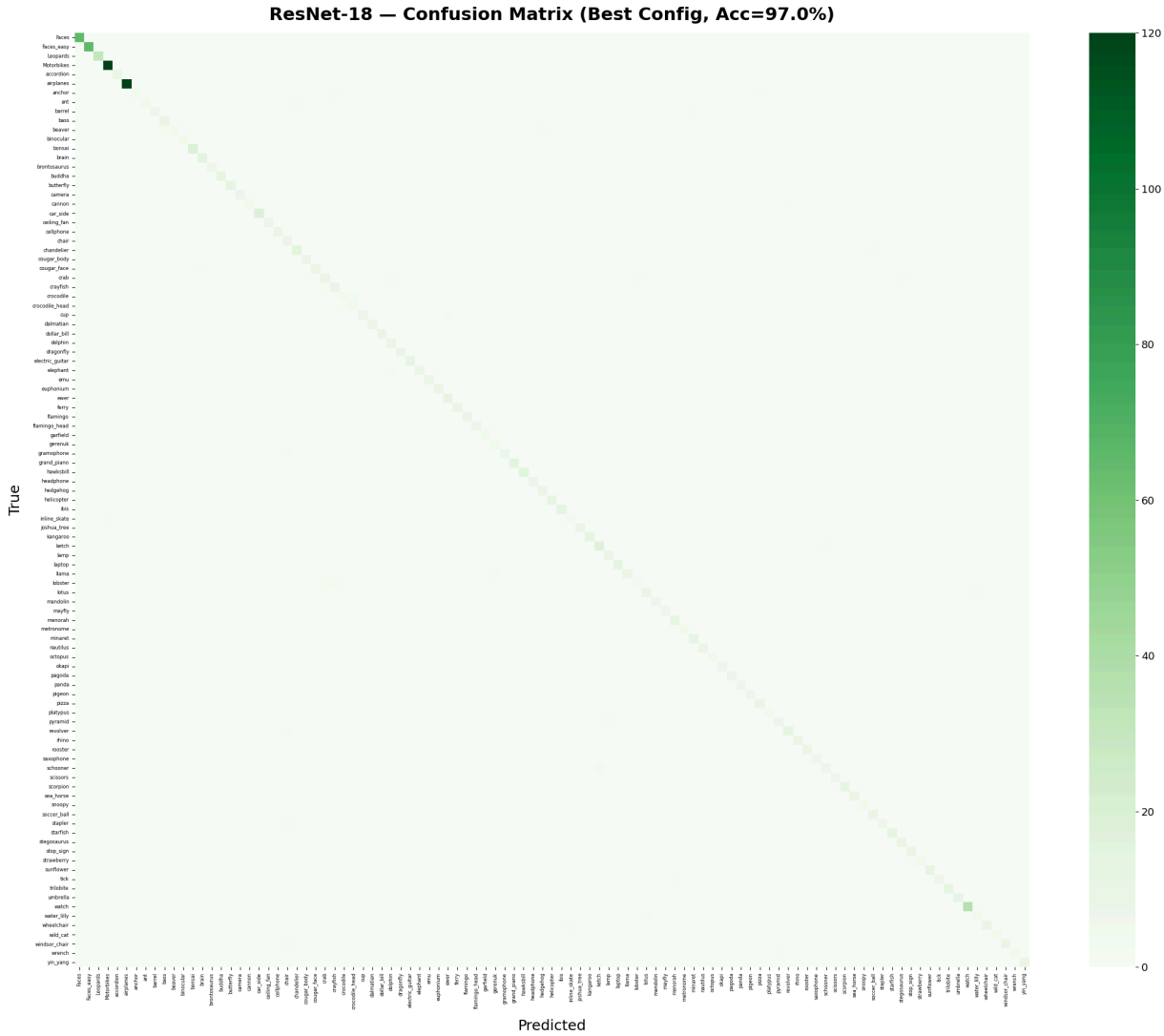
In [16]:

```
# Confusion Matrix for Best Config of Each Model

MODEL_LABELS = {"svm": "SVM", "resnet18": "ResNet-18",
                "efficientnet_b0": "EfficientNet-B0", "vit_small": "ViT-Smal

for model_name in ["svm", "resnet18", "efficientnet_b0", "vit_small"]:
    subset = results_df[results_df["model"] == model_name]
    if subset.empty:
        continue

    best_key = subset.iloc[0]["run_key"]
    cm_data  = np.array(all_results[best_key]["confusion_matrix"])

    fig, ax = plt.subplots(figsize=(22, 18))

    cmap = {"svm": "Blues", "resnet18": "Greens",
            "efficientnet_b0": "Oranges", "vit_small": "Purples"}

    sns.heatmap(cm_data, annot=False, fmt="d",
                cmap=cmap.get(model_name, "Blues"),
                xticklabels=le.classes_,
                yticklabels=le.classes_, ax=ax)

    ax.set_xlabel("Predicted", fontsize=18, labelpad=12)
    ax.set_ylabel("True",      fontsize=18, labelpad=12)

    acc = subset.iloc[0]["test_acc_%"]
    ax.set_title(
        f"{MODEL_LABELS[model_name]} — Confusion Matrix (Best Config, Acc={a
        fontsize=22, fontweight="bold", pad=16
    )

    ax.tick_params(axis="x", labelsize=6, rotation=90)
    ax.tick_params(axis="y", labelsize=6, rotation=0)

    # Colorbar label size
    cbar = ax.collections[0].colorbar
    cbar.ax.tick_params(labelsize=13)
```

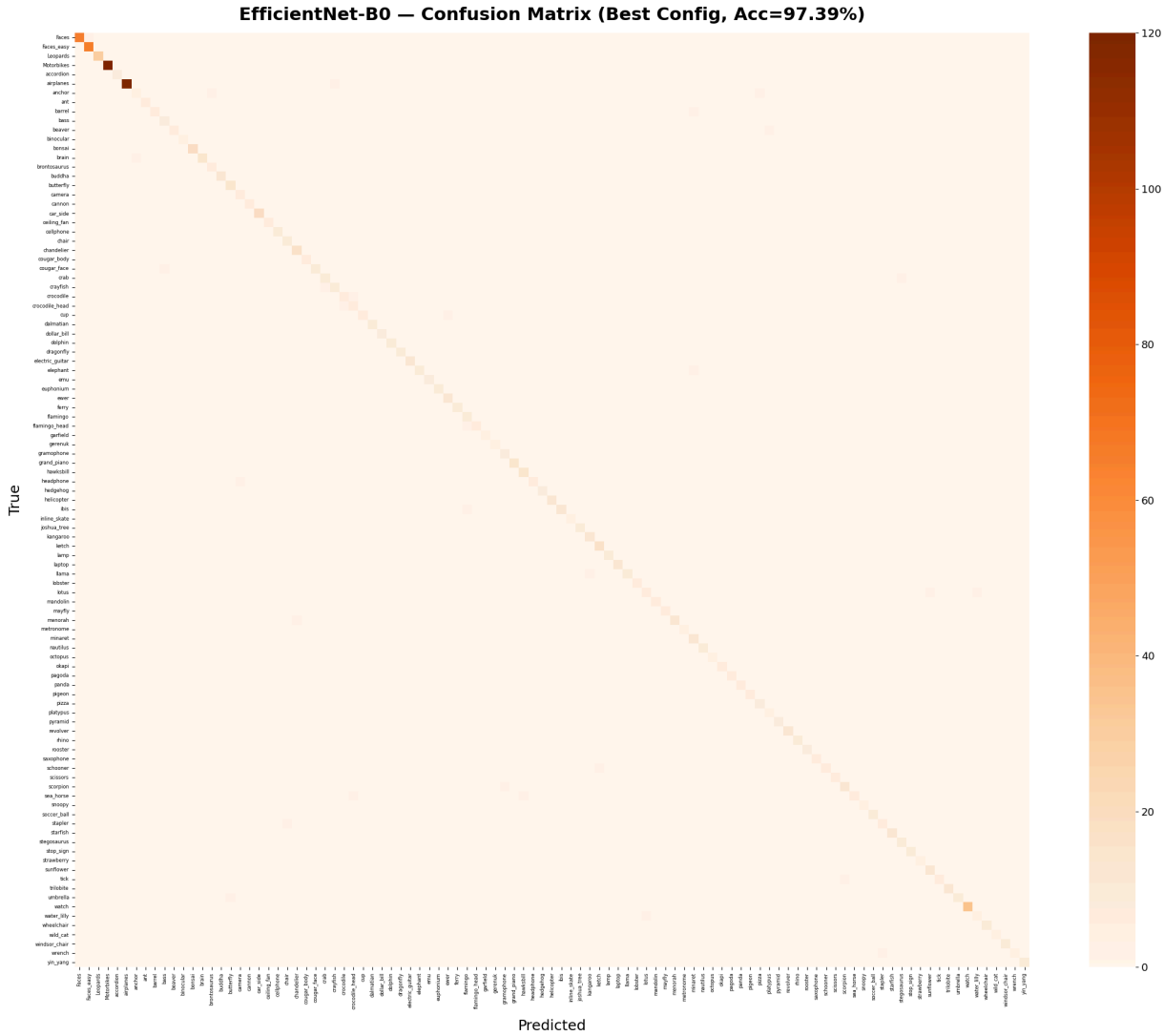
```
    plt.tight_layout()
    plt.savefig(FIG_DIR / f"{model_name}_best_confusion_matrix.png", dpi=150
    plt.show()
    print(f"Saved confusion matrix for {model_name} ✓")
```



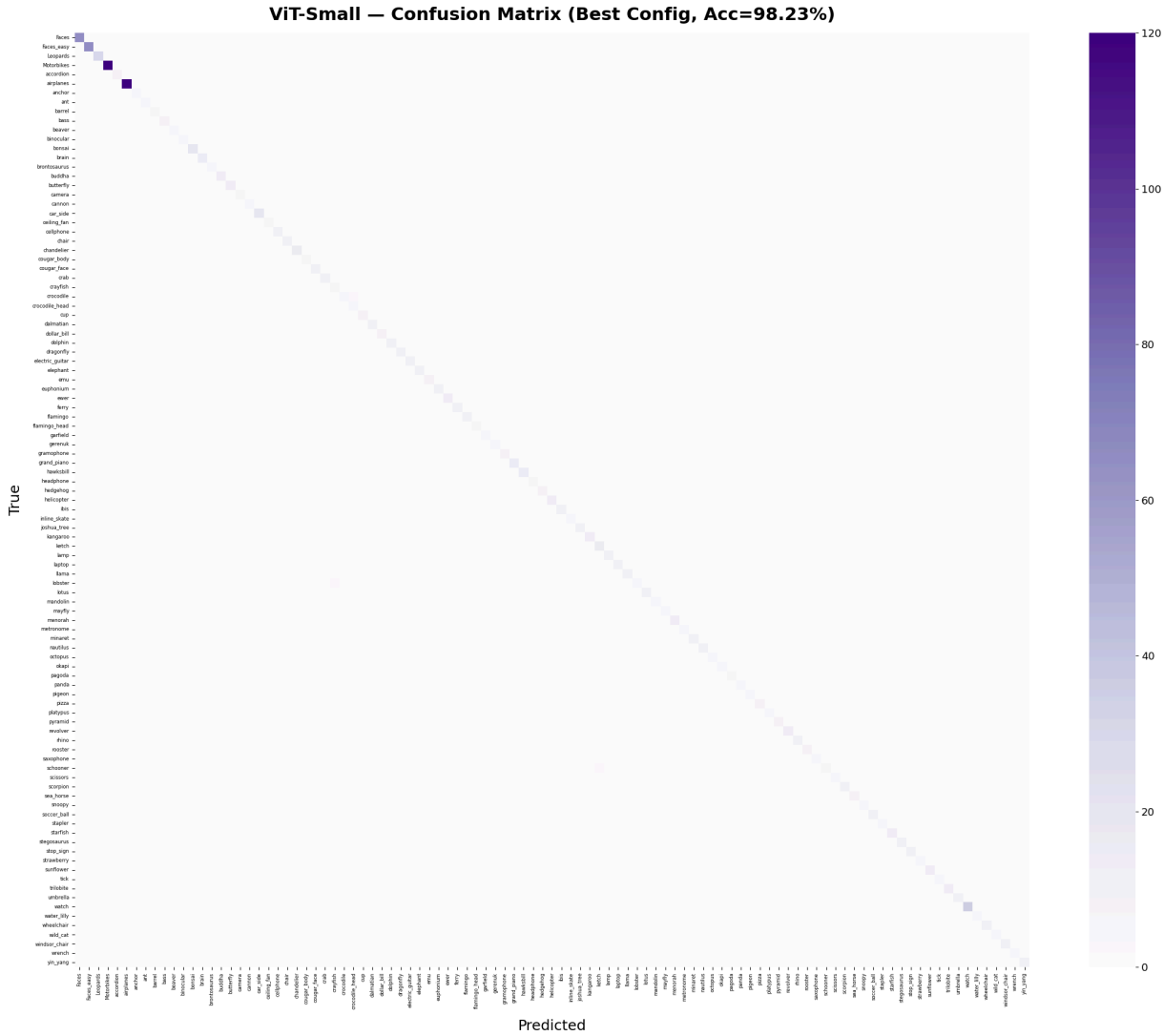

```
Saved confusion matrix for svm ✓
```




```
Saved confusion matrix for resnet18 ✓
```




```
Saved confusion matrix for efficientnet_b0 ✓
```

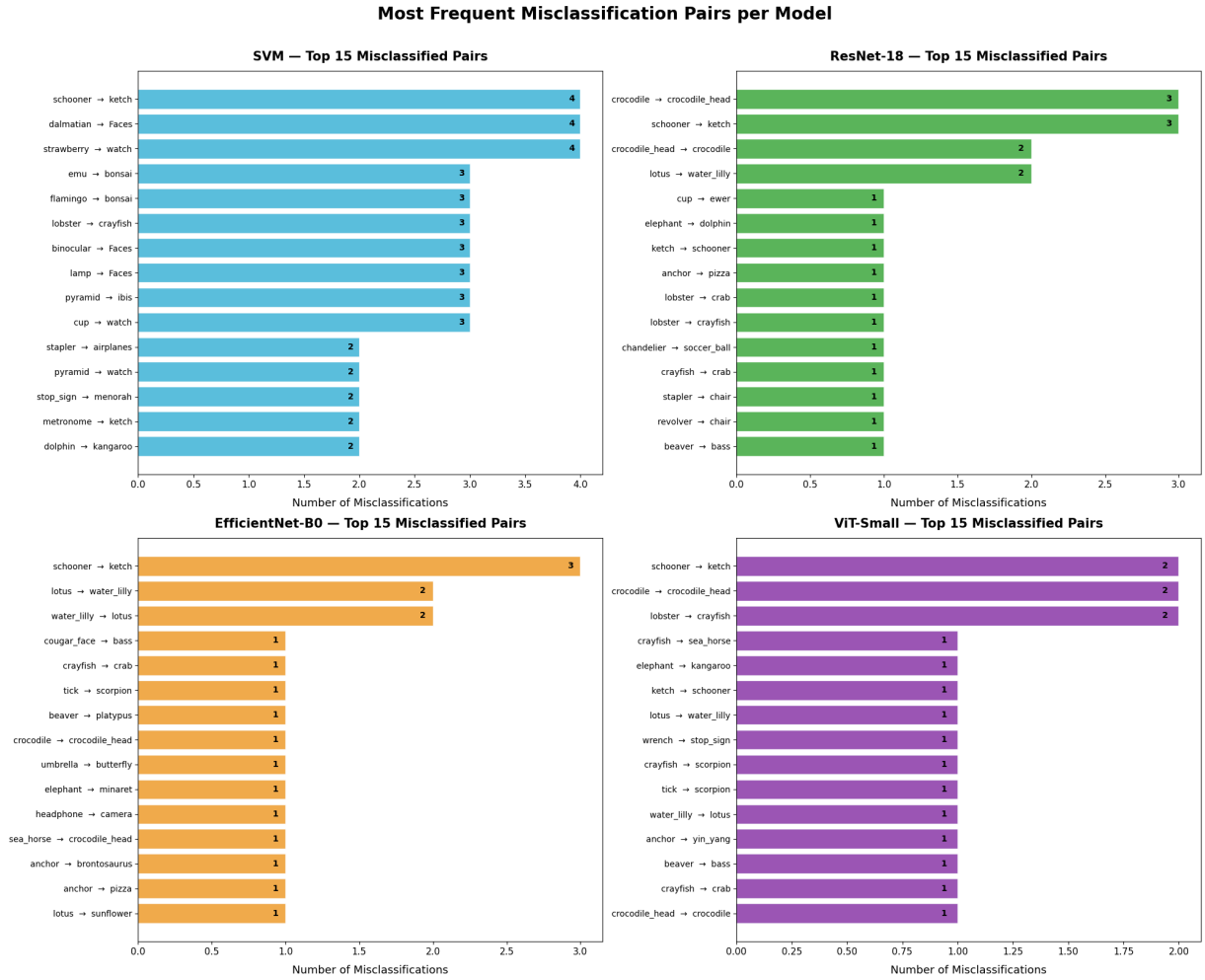



```
Saved confusion matrix for vit_small ✓
```

In [17]:

```
# Show the most frequent specific misclassifications as a horizontal bar cha

# Top-N Misclassified Pairs

color_map = {"svm": "#5bc0de", "resnet18": "#5cb85c",
             "efficientnet_b0": "#f0ad4e", "vit_small": "#9b59b6"}

MODEL_LABELS = {"svm": "SVM", "resnet18": "ResNet-18",
                "efficientnet_b0": "EfficientNet-B0", "vit_small": "ViT-Smal

fig, axes = plt.subplots(2, 2, figsize=(20, 16))
axes = axes.flatten()

for i, model_name in enumerate(["svm", "resnet18", "efficientnet_b0", "vit_s
    subset = results_df[results_df["model"] == model_name]
    if subset.empty:
        continue

    cm_data = np.array(all_results[subset.iloc[0]["run_key"]]["confusion_mat
    np.fill_diagonal(cm_data, 0)  # zero out diagonal to focus on errors onl

    # Get top 15 misclassified pairs
```

```
    flat_idx    = np.argsort(cm_data.flatten())[::-1][:15]
    true_idx    = flat_idx // cm_data.shape[0]
    pred_idx    = flat_idx  % cm_data.shape[0]
    counts      = cm_data[true_idx, pred_idx]
    pair_labels = [f"{le.classes_[t]}  →  {le.classes_[p]}"
                   for t, p in zip(true_idx, pred_idx)]

    color = color_map.get(model_name, "gray")
    axes[i].barh(range(len(counts))[::-1], counts, color=color, edgecolor="w

    for j, (val, label) in enumerate(zip(counts, pair_labels)):
        axes[i].text(val - 0.05, len(counts) - 1 - j, f"{val}",
                     va="center", ha="right", fontsize=10, fontweight="bold"

    axes[i].set_yticks(range(len(pair_labels)))
    axes[i].set_yticklabels(pair_labels[::-1], fontsize=10)
    axes[i].set_xlabel("Number of Misclassifications", fontsize=13, labelpad
    axes[i].set_title(
        f"{MODEL_LABELS[model_name]} — Top 15 Misclassified Pairs",
        fontsize=15, fontweight="bold", pad=12
    )
    axes[i].tick_params(axis="x", labelsize=11)

plt.suptitle("Most Frequent Misclassification Pairs per Model",
             fontsize=20, fontweight="bold", y=1.01)
plt.tight_layout()
plt.savefig(FIG_DIR / "top_misclassified_pairs.png", dpi=150, bbox_inches="t
plt.show()
print("Saved: top_misclassified_pairs.png ✓")
```

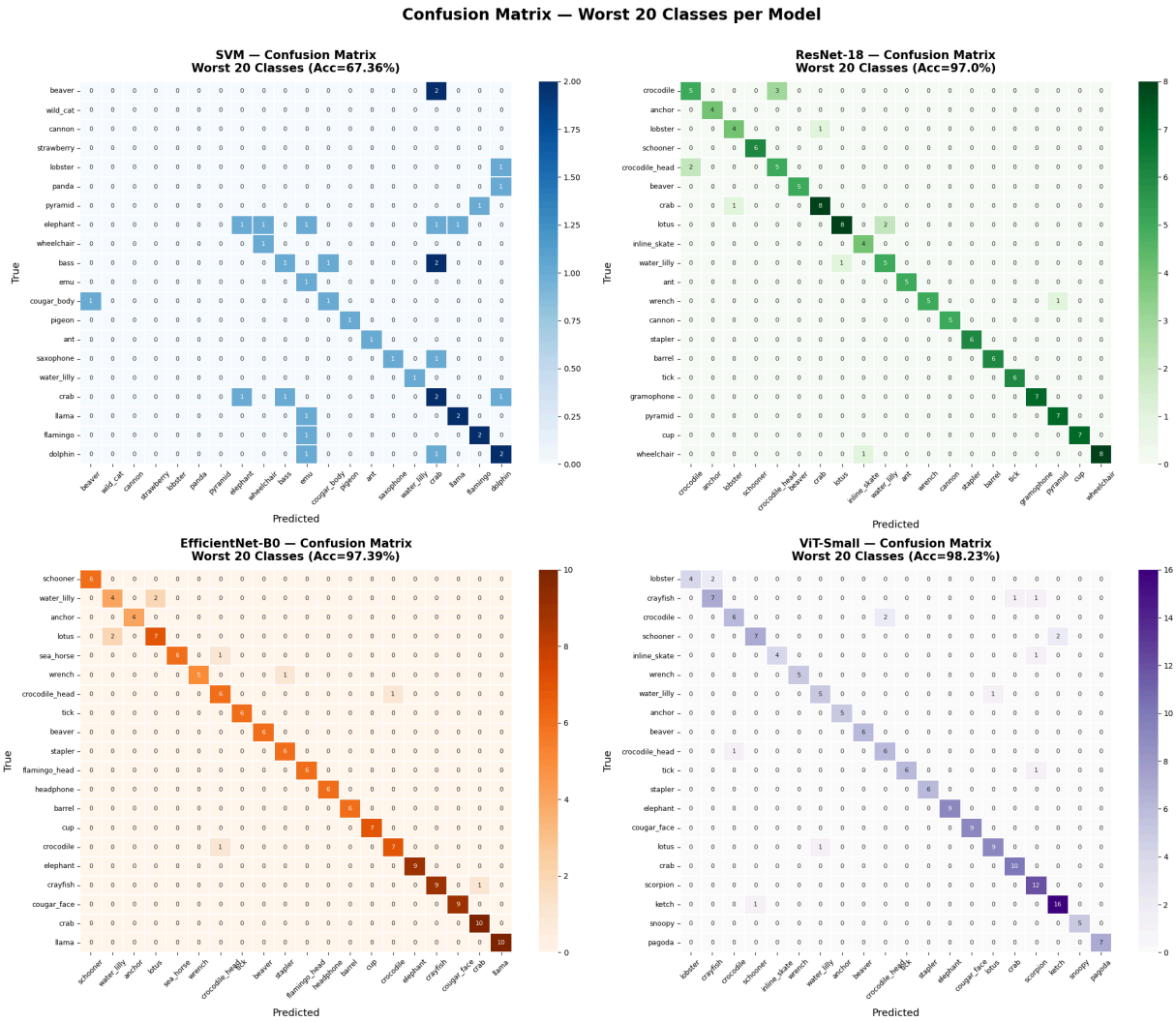



```
Saved: top_misclassified_pairs.png ✓
```

In [18]:

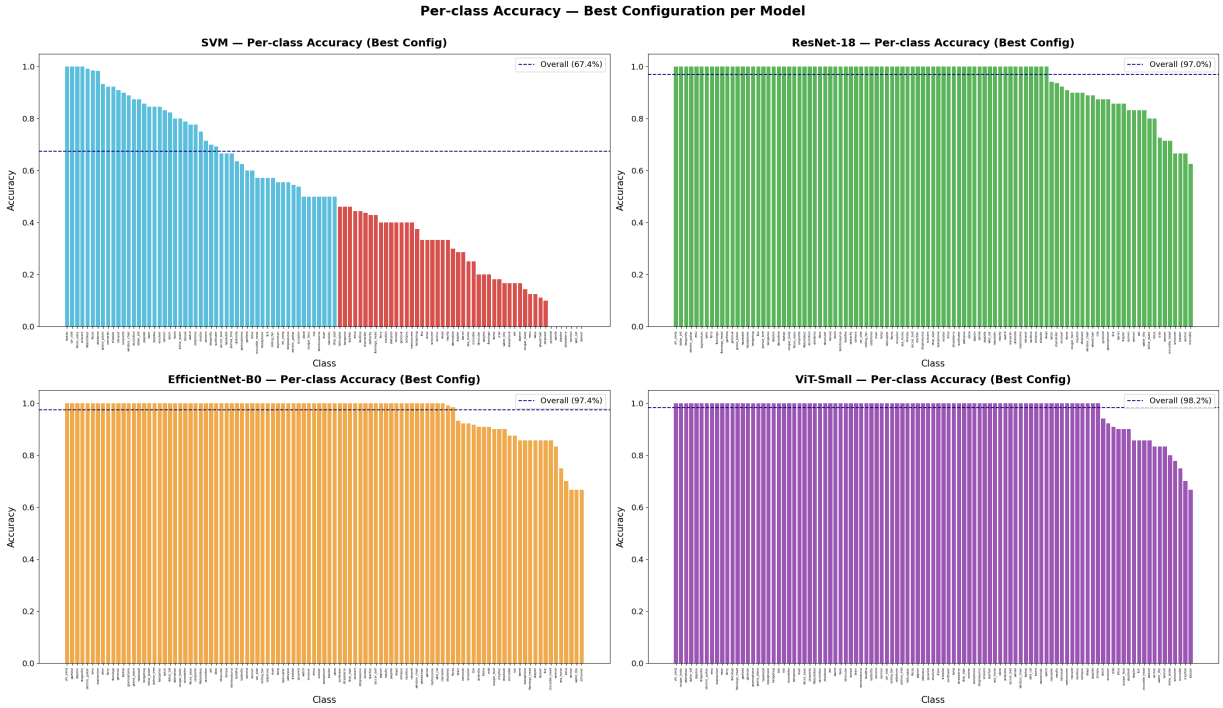
```python
# Zoom into only the worst-performing classes
N = 20  # show only bottom 20 classes by accuracy

fig, axes = plt.subplots(2, 2, figsize=(22, 18))
axes = axes.flatten()

cmap = {"svm": "Blues", "resnet18": "Greens",
        "efficientnet_b0": "Oranges", "vit_small": "Purples"}

for i, model_name in enumerate(["svm", "resnet18", "efficientnet_b0", "vit_s
    subset = results_df[results_df["model"] == model_name]
    if subset.empty:
        continue

    best_key = subset.iloc[0]["run_key"]
    cm_data  = np.array(all_results[best_key]["confusion_matrix"])
    pca      = np.array(all_results[best_key]["per_class_acc"])

    # Select N worst classes by per-class accuracy
    worst_idx   = np.argsort(pca)[:N]
    cm_subset   = cm_data[np.ix_(worst_idx, worst_idx)]
    class_names = le.classes_[worst_idx]

    sns.heatmap(cm_subset, annot=True, fmt="d",
                cmap=cmap.get(model_name, "Blues"),
```

```
                xticklabels=class_names,
                yticklabels=class_names,
                ax=axes[i], linewidths=0.3,
                annot_kws={"size": 8})

    acc = subset.iloc[0]["test_acc_%"]
    axes[i].set_title(
        f"{MODEL_LABELS[model_name]} — Confusion Matrix\nWorst {N} Classes (
        fontsize=15, fontweight="bold", pad=12
    )
    axes[i].set_xlabel("Predicted", fontsize=13, labelpad=10)
    axes[i].set_ylabel("True",      fontsize=13, labelpad=10)
    axes[i].tick_params(axis="x", labelsize=9, rotation=45)
    axes[i].tick_params(axis="y", labelsize=9, rotation=0)

plt.suptitle(f"Confusion Matrix — Worst {N} Classes per Model",
             fontsize=20, fontweight="bold", y=1.01)
plt.tight_layout()
plt.savefig(FIG_DIR / "confusion_matrix_worst_classes.png", dpi=150, bbox_ir
plt.show()
print("Saved: confusion_matrix_worst_classes.png ✓")
```




```
Saved: confusion_matrix_worst_classes.png ✓
```

In [19]:
```
# Per-class Accuracy for Best Config of Each Model

MODEL_LABELS = {"svm": "SVM", "resnet18": "ResNet-18",
                "efficientnet_b0": "EfficientNet-B0", "vit_small": "ViT-Smal

fig, axes = plt.subplots(2, 2, figsize=(28, 16))
axes = axes.flatten()

color_map = {"svm": "#5bc0de", "resnet18": "#5cb85c",
             "efficientnet_b0": "#f0ad4e", "vit_small": "#9b59b6"}

for i, model_name in enumerate(["svm", "resnet18", "efficientnet_b0", "vit_s
    subset = results_df[results_df["model"] == model_name]
    if subset.empty:
        axes[i].set_title(f"{MODEL_LABELS[model_name]} — no results", fontsi
        continue

    best_key    = subset.iloc[0]["run_key"]
    pca         = np.array(all_results[best_key]["per_class_acc"])
    overall_acc = subset.iloc[0]["test_acc_%"] / 100

    sorted_idx   = np.argsort(pca)[::-1]
    sorted_pca   = pca[sorted_idx]
    sorted_names = le.classes_[sorted_idx]

    color      = color_map.get(model_name, "gray")
    bar_colors = ["#d9534f" if v < 0.5 else color for v in sorted_pca]

    axes[i].bar(range(len(sorted_pca)), sorted_pca, color=bar_colors)
    axes[i].axhline(overall_acc, color="navy", linestyle="--",
                    label=f"Overall ({overall_acc*100:.1f}%)")

    axes[i].set_title(
        f"{MODEL_LABELS[model_name]} — Per-class Accuracy (Best Config)",
        fontsize=18, fontweight="bold", pad=12
    )
    axes[i].set_ylabel("Accuracy", fontsize=15, labelpad=10)
    axes[i].set_xlabel("Class", fontsize=15, labelpad=10)

    axes[i].set_xticks(range(len(sorted_names)))
    axes[i].set_xticklabels(sorted_names, rotation=90, fontsize=5)
    axes[i].tick_params(axis="y", labelsize=13)

    axes[i].legend(fontsize=13)

plt.suptitle("Per-class Accuracy — Best Configuration per Model",
             fontsize=22, fontweight="bold", y=1.01)
plt.tight_layout()
plt.savefig(FIG_DIR / "all_models_per_class_accuracy.png", dpi=150, bbox_inc
plt.show()
print("Saved: all_models_per_class_accuracy.png ✓")
```

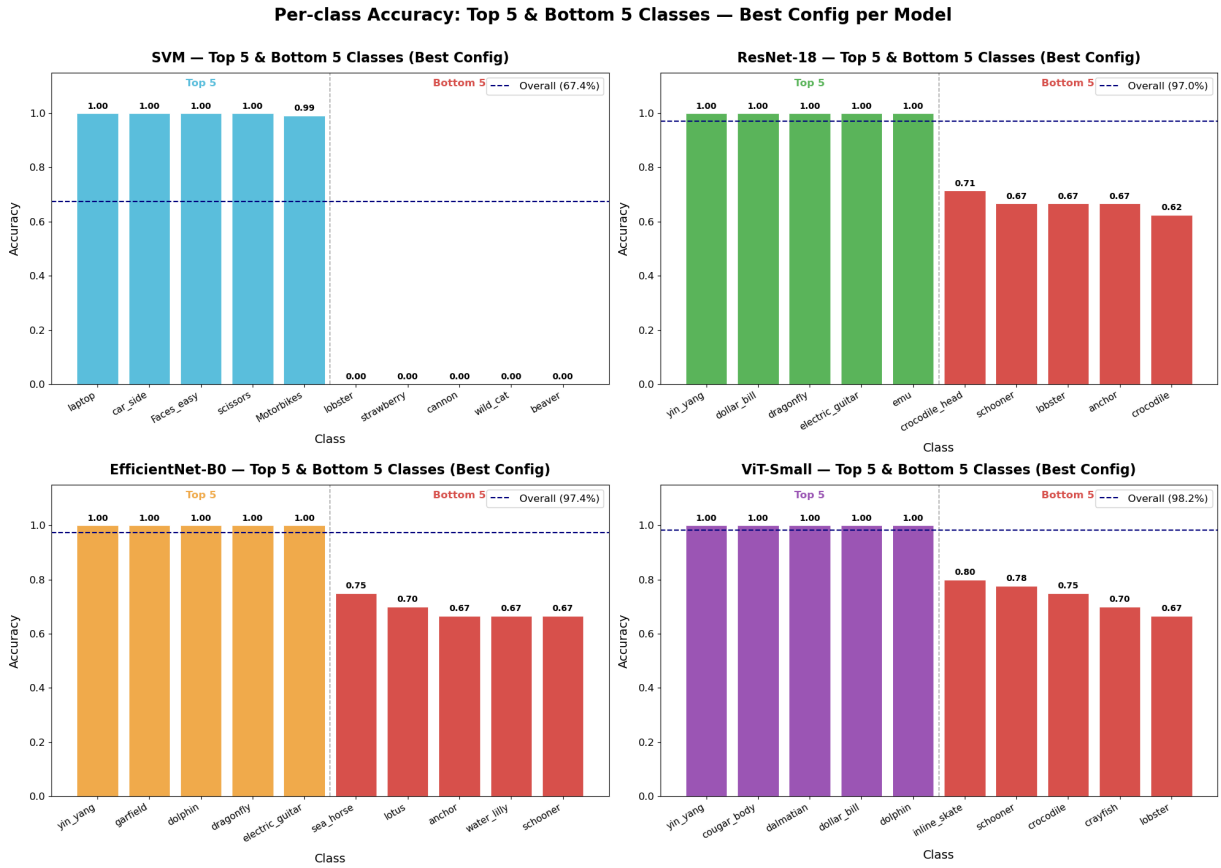



```
Saved: all_models_per_class_accuracy.png ✓
```

In [20]:

```python
# Per-class Accuracy — Top 5 and Bottom 5 per Model

MODEL_LABELS = {"svm": "SVM", "resnet18": "ResNet-18",
                "efficientnet_b0": "EfficientNet-B0", "vit_small": "ViT-Smal

fig, axes = plt.subplots(2, 2, figsize=(20, 14))
axes = axes.flatten()

color_map = {"svm": "#5bc0de", "resnet18": "#5cb85c",
             "efficientnet_b0": "#f0ad4e", "vit_small": "#9b59b6"}

for i, model_name in enumerate(["svm", "resnet18", "efficientnet_b0", "vit_s
    subset = results_df[results_df["model"] == model_name]
    if subset.empty:
        axes[i].set_title(f"{MODEL_LABELS[model_name]} — no results", fontsi
        continue

    best_key    = subset.iloc[0]["run_key"]
    pca         = np.array(all_results[best_key]["per_class_acc"])
    overall_acc = subset.iloc[0]["test_acc_%"] / 100

    sorted_idx   = np.argsort(pca)[::-1]
    sorted_pca   = pca[sorted_idx]
    sorted_names = le.classes_[sorted_idx]

    # Top 5 and bottom 5
    top_idx    = list(range(5))
    bot_idx    = list(range(len(sorted_pca) - 5, len(sorted_pca)))
    sel_idx    = top_idx + bot_idx
    sel_pca    = sorted_pca[sel_idx]
    sel_names  = sorted_names[sel_idx]

    color      = color_map.get(model_name, "gray")
```

```
    bar_colors = (
        [color] * 5 +          # top 5 — model color
        ["#d9534f"] * 5        # bottom 5 — red
    )

    x = np.arange(len(sel_names))
    bars = axes[i].bar(x, sel_pca, color=bar_colors, edgecolor="white", line

    # Value annotations on each bar
    for bar, val in zip(bars, sel_pca):
        axes[i].text(
            bar.get_x() + bar.get_width() / 2,
            val + 0.01,
            f"{val:.2f}",
            ha="center", va="bottom", fontsize=10, fontweight="bold"
        )

    # Divider line between top 5 and bottom 5
    axes[i].axvline(x=4.5, color="gray", linestyle="--", linewidth=1.2, alph

    # Overall accuracy reference line
    axes[i].axhline(overall_acc, color="navy", linestyle="--", linewidth=1.5
                    label=f"Overall ({overall_acc*100:.1f}%)")

    axes[i].set_title(
        f"{MODEL_LABELS[model_name]} — Top 5 & Bottom 5 Classes (Best Config
        fontsize=16, fontweight="bold", pad=12
    )
    axes[i].set_ylabel("Accuracy", fontsize=14, labelpad=10)
    axes[i].set_xlabel("Class", fontsize=14, labelpad=10)
    axes[i].set_ylim(0, 1.15)
    axes[i].set_xticks(x)
    axes[i].set_xticklabels(sel_names, rotation=30, ha="right", fontsize=11)
    axes[i].tick_params(axis="y", labelsize=12)
    axes[i].legend(fontsize=12)

    # Top / Bottom group labels
    axes[i].text(2,   1.10, "Top 5",    ha="center", fontsize=12, color=colo
    axes[i].text(7,   1.10, "Bottom 5", ha="center", fontsize=12, color="#d9

plt.suptitle("Per-class Accuracy: Top 5 & Bottom 5 Classes — Best Config per
             fontsize=20, fontweight="bold", y=1.01)
plt.tight_layout()
plt.savefig(FIG_DIR / "top_bottom5_per_class_accuracy.png", dpi=150, bbox_in
plt.show()
print("Saved: top_bottom5_per_class_accuracy.png ✓")
```




```
Saved: top_bottom5_per_class_accuracy.png ✓
```

In [21]:

```python
# Ablation Summary Plots

DL_ORDER    = ["resnet18", "efficientnet_b0", "vit_small"]
ALL_ORDER   = ["svm", "resnet18", "efficientnet_b0", "vit_small"]
MODEL_LABELS = {
    "svm": "SVM", "resnet18": "ResNet-18",
    "efficientnet_b0": "EfficientNet-B0", "vit_small": "ViT-Small"
}

TITLE_SIZE  = 16
LABEL_SIZE  = 14
TICK_SIZE   = 13
LEGEND_SIZE = 12

def annotate_bars(ax):
    """Add value labels on top of every bar in a grouped bar chart."""
    for bar in ax.patches:
        val = bar.get_height()
        if val > 0:
            ax.text(
                bar.get_x() + bar.get_width() / 2,
                val + 0.3,
                f"{val:.1f}%",
                ha="center", va="bottom",
                fontsize=9, fontweight="bold"
            )
    # add headroom so labels aren't clipped
    ax.set_ylim(0, ax.get_ylim()[1] + 6)
```

```
fig, axes = plt.subplots(2, 2, figsize=(20, 14))

# Ablation 1: Image Size (DL only)
ax = axes[0, 0]
img_abl = (results_df[results_df["model"] != "svm"]
           .groupby(["model", "img_size"])["test_acc_%"].mean().unstack())
img_abl.index = pd.CategoricalIndex(img_abl.index, categories=DL_ORDER, orde
img_abl = img_abl.sort_index()
img_abl.index = [MODEL_LABELS[m] for m in img_abl.index]
img_abl.plot(kind="bar", ax=ax, colormap="coolwarm")
annotate_bars(ax)
ax.set_title("Ablation: Image Size (avg across other params)", fontsize=TITL
ax.set_ylabel("Avg Test Accuracy (%)", fontsize=LABEL_SIZE, labelpad=10)
ax.set_xlabel("Model", fontsize=LABEL_SIZE, labelpad=10)
ax.tick_params(axis='x', rotation=15, labelsize=TICK_SIZE)
ax.tick_params(axis='y', labelsize=TICK_SIZE)
ax.legend(title="Image Size", fontsize=LEGEND_SIZE, title_fontsize=LEGEND_SI

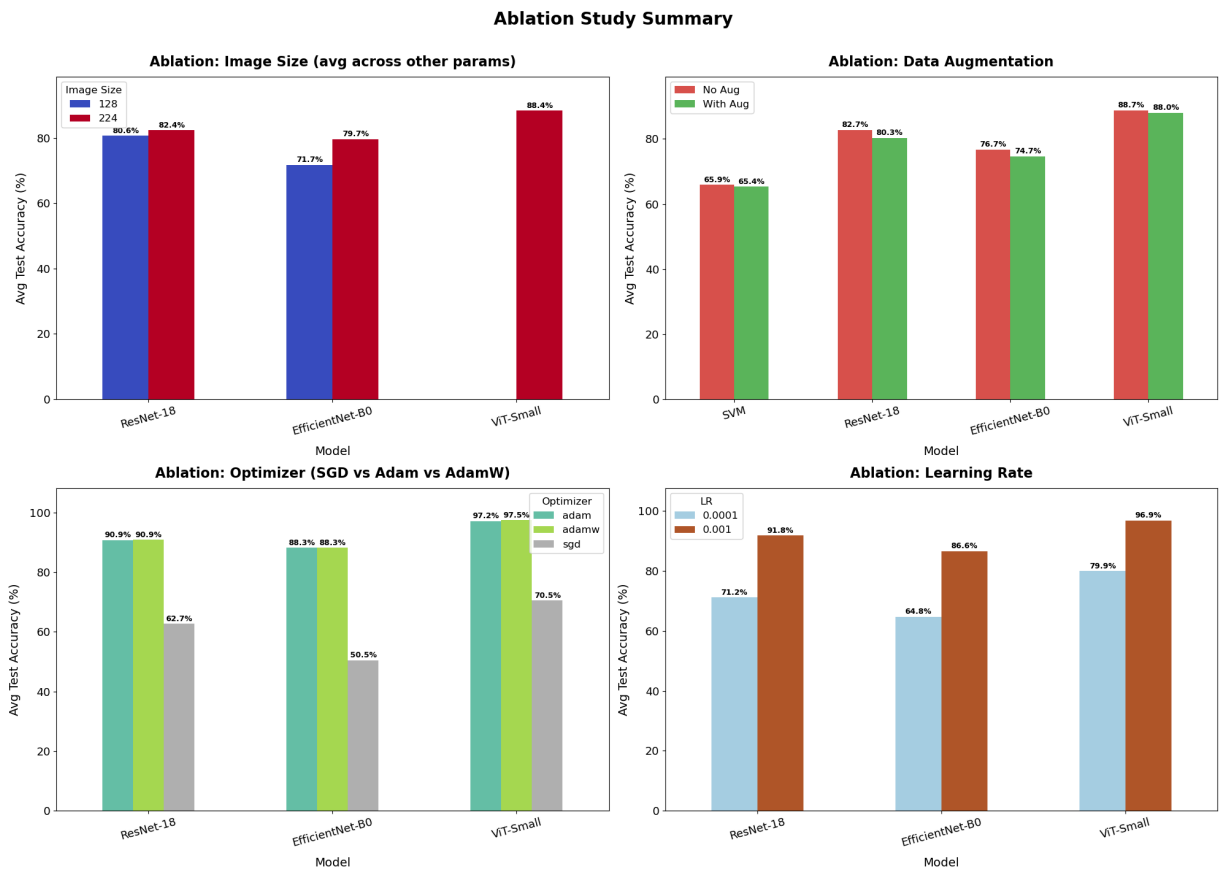
# Ablation 2: Augmentation (all models)
ax = axes[0, 1]
aug_abl = results_df.groupby(["model", "augment"])["test_acc_%"].mean().unst
aug_abl.index = pd.CategoricalIndex(aug_abl.index, categories=ALL_ORDER, orc
aug_abl = aug_abl.sort_index()
aug_abl.columns = ["No Aug", "With Aug"]
aug_abl.index = [MODEL_LABELS[m] for m in aug_abl.index]
aug_abl.plot(kind="bar", ax=ax, color=["#d9534f", "#5cb85c"])
annotate_bars(ax)
ax.set_title("Ablation: Data Augmentation", fontsize=TITLE_SIZE, fontweight=
ax.set_ylabel("Avg Test Accuracy (%)", fontsize=LABEL_SIZE, labelpad=10)
ax.set_xlabel("Model", fontsize=LABEL_SIZE, labelpad=10)
ax.tick_params(axis='x', rotation=15, labelsize=TICK_SIZE)
ax.tick_params(axis='y', labelsize=TICK_SIZE)
ax.legend(fontsize=LEGEND_SIZE)

# Ablation 3: Optimizer (DL only)
ax = axes[1, 0]
dl_only = results_df[results_df["model"] != "svm"]
opt_abl = dl_only.groupby(["model", "optimizer"])["test_acc_%"].mean().unsta
opt_abl.index = pd.CategoricalIndex(opt_abl.index, categories=DL_ORDER, orde
opt_abl = opt_abl.sort_index()
opt_abl.index = [MODEL_LABELS[m] for m in opt_abl.index]
opt_abl.plot(kind="bar", ax=ax, colormap="Set2")
annotate_bars(ax)
ax.set_title("Ablation: Optimizer (SGD vs Adam vs AdamW)", fontsize=TITLE_SI
ax.set_ylabel("Avg Test Accuracy (%)", fontsize=LABEL_SIZE, labelpad=10)
ax.set_xlabel("Model", fontsize=LABEL_SIZE, labelpad=10)
ax.tick_params(axis='x', rotation=15, labelsize=TICK_SIZE)
ax.tick_params(axis='y', labelsize=TICK_SIZE)
ax.legend(title="Optimizer", fontsize=LEGEND_SIZE, title_fontsize=LEGEND_SIZ

# Ablation 4: Learning Rate (DL only)
ax = axes[1, 1]
lr_abl = dl_only.groupby(["model", "lr"])["test_acc_%"].mean().unstack()
lr_abl.index = pd.CategoricalIndex(lr_abl.index, categories=DL_ORDER, ordere
lr_abl = lr_abl.sort_index()
```

```
lr_abl.index = [MODEL_LABELS[m] for m in lr_abl.index]
lr_abl.plot(kind="bar", ax=ax, colormap="Paired")
annotate_bars(ax)
ax.set_title("Ablation: Learning Rate", fontsize=TITLE_SIZE, fontweight="bol
ax.set_ylabel("Avg Test Accuracy (%)", fontsize=LABEL_SIZE, labelpad=10)
ax.set_xlabel("Model", fontsize=LABEL_SIZE, labelpad=10)
ax.tick_params(axis='x', rotation=15, labelsize=TICK_SIZE)
ax.tick_params(axis='y', labelsize=TICK_SIZE)
ax.legend(title="LR", fontsize=LEGEND_SIZE, title_fontsize=LEGEND_SIZE)

plt.suptitle("Ablation Study Summary", fontsize=20, fontweight="bold", y=1.0
plt.tight_layout()
plt.savefig(FIG_DIR / "ablation_summary.png", dpi=150, bbox_inches="tight")
plt.show()
print("Saved: ablation_summary.png ✓")
```




```
Saved: ablation_summary.png ✓
```

In [22]:

```
# Export Final Summary CSV
results_df.to_csv(RESULTS_DIR / "summary_table.csv", index=False)

print("Files saved to RESULTS_DIR:")
for f in sorted(RESULTS_DIR.iterdir()):
    print(f"  {f.name}")

print("\nFiles saved to FIG_DIR:")
for f in sorted(FIG_DIR.iterdir()):
    print(f"  {f.name}")

print("\nFull experiment grid complete ✓")
```

```
print(f"Best overall model: {results_df.iloc[0]['run_key']}")
print(f"Best accuracy     : {results_df.iloc[0]['test_acc_%']}%")
```

```
Files saved to RESULTS_DIR:
  all_results.json
  summary_table.csv

Files saved to FIG_DIR:
  ablation_summary.png
  all_models_per_class_accuracy.png
  comparison_summary.png
  confusion_matrix_worst_classes.png
  efficientnet_b0_best_confusion_matrix.png
  efficientnet_b0_img128_ep10_lr0.0001_adam_augFalse_curves.png
  efficientnet_b0_img128_ep10_lr0.0001_adam_augTrue_curves.png
  efficientnet_b0_img128_ep10_lr0.0001_adamw_augFalse_curves.png
  efficientnet_b0_img128_ep10_lr0.0001_adamw_augTrue_curves.png
  efficientnet_b0_img128_ep10_lr0.0001_sgd_augFalse_curves.png
  efficientnet_b0_img128_ep10_lr0.0001_sgd_augTrue_curves.png
  efficientnet_b0_img128_ep10_lr0.001_adam_augFalse_curves.png
  efficientnet_b0_img128_ep10_lr0.001_adam_augTrue_curves.png
  efficientnet_b0_img128_ep10_lr0.001_adamw_augFalse_curves.png
  efficientnet_b0_img128_ep10_lr0.001_adamw_augTrue_curves.png
  efficientnet_b0_img128_ep10_lr0.001_sgd_augFalse_curves.png
  efficientnet_b0_img128_ep10_lr0.001_sgd_augTrue_curves.png
  efficientnet_b0_img128_ep5_lr0.0001_adam_augFalse_curves.png
  efficientnet_b0_img128_ep5_lr0.0001_adam_augTrue_curves.png
  efficientnet_b0_img128_ep5_lr0.0001_adamw_augFalse_curves.png
  efficientnet_b0_img128_ep5_lr0.0001_adamw_augTrue_curves.png
  efficientnet_b0_img128_ep5_lr0.0001_sgd_augFalse_curves.png
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  efficientnet_b0_img128_ep5_lr0.001_adamw_augTrue_curves.png
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  efficientnet_b0_img224_ep10_lr0.001_sgd_augFalse_curves.png
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  efficientnet_b0_img224_ep5_lr0.0001_adam_augFalse_curves.png
  efficientnet_b0_img224_ep5_lr0.0001_adam_augTrue_curves.png
  efficientnet_b0_img224_ep5_lr0.0001_adamw_augFalse_curves.png
  efficientnet_b0_img224_ep5_lr0.0001_adamw_augTrue_curves.png
  efficientnet_b0_img224_ep5_lr0.0001_sgd_augFalse_curves.png
  efficientnet_b0_img224_ep5_lr0.0001_sgd_augTrue_curves.png
  efficientnet_b0_img224_ep5_lr0.001_adam_augFalse_curves.png
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  efficientnet_b0_img224_ep5_lr0.001_adamw_augFalse_curves.png
  efficientnet_b0_img224_ep5_lr0.001_adamw_augTrue_curves.png
```

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resnet18_best_confusion_matrix.png
resnet18_img128_ep10_lr0.0001_adam_augFalse_curves.png
resnet18_img128_ep10_lr0.0001_adam_augTrue_curves.png
resnet18_img128_ep10_lr0.0001_adamw_augFalse_curves.png
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resnet18_img128_ep5_lr0.0001_adam_augTrue_curves.png
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resnet18_img224_ep10_lr0.001_adamw_augFalse_curves.png
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resnet18_img224_ep5_lr0.001_sgd_augTrue_curves.png
svm_best_confusion_matrix.png
top_bottom5_per_class_accuracy.png
top_misclassified_pairs.png
vit_small_best_confusion_matrix.png
vit_small_img224_ep10_lr0.0001_adam_augFalse_curves.png
```

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vit_small_img224_ep10_lr0.0001_adam_augTrue_curves.png
vit_small_img224_ep10_lr0.0001_adamw_augFalse_curves.png
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vit_small_img224_ep5_lr0.001_sgd_augFalse_curves.png
vit_small_img224_ep5_lr0.001_sgd_augTrue_curves.png

Full experiment grid complete ✓
Best overall model: vit_small_img224_ep10_lr0.001_adamw_augFalse
Best accuracy     : 98.23%
```

In [23]:

```
import pandas as pd
from IPython.display import display, HTML

# Build display table from all_results
rows = []
for key, result in all_results.items():
    if "error" in result:
        continue

    cfg = result["config"]
    m   = result["metrics"]

    # Format config string
    if cfg["model"] == "svm":
        config_str = f"img={cfg['img_size']}, C={cfg['C']}, aug={cfg['augmer
    else:
        config_str = (f"img={cfg['img_size']}, ep={cfg['epochs']}, "
                      f"lr={cfg['lr']}, opt={cfg['optimizer']}, "
                      f"aug={cfg['augment']}")

    rows.append({
        "Model":            cfg["model"],
        "Accuracy (%)":     round(m["accuracy"]      * 100, 2),
        "Top-5 Accuracy (%)": round(m["top5_accuracy"] * 100, 2),
        "Macro F1 Score":   round(m["macro_f1"],          3),
        "Configuration":    config_str,
        "Train Time (min)": round(result["train_time_sec"] / 60, 1),
    })
```

```
display_df = (pd.DataFrame(rows)
              .sort_values(["Model", "Accuracy (%)"], ascending=[True, False
              .reset_index(drop=True))

# Clean model name labels
MODEL_LABELS = {
    "svm":            "SVM",
    "resnet18":       "ResNet-18",
    "efficientnet_b0":"EfficientNet-B0",
    "vit_small":      "ViT-Small",
}
display_df["Model"] = display_df["Model"].map(MODEL_LABELS)

print(f"Total runs: {len(display_df)}\n")

# ── Styled HTML table ──────────────────────────────────────────────────
MODEL_COLORS = {
    "SVM":             "#d0ecf7",
    "ResNet-18":       "#d4edda",
    "EfficientNet-B0": "#fff3cd",
    "ViT-Small":       "#e8d5f5",
}

def highlight_rows(row):
    color = MODEL_COLORS.get(row["Model"], "#ffffff")
    return [f"background-color: {color}"] * len(row)

styled = (display_df.style
          .apply(highlight_rows, axis=1)
          .format({
              "Accuracy (%)":       "{:.2f}%",
              "Top-5 Accuracy (%)": "{:.2f}%",
              "Macro F1 Score":     "{:.3f}",
              "Train Time (min)":   "{:.1f}",
          })
          .set_table_styles([
              {"selector": "thead th",
               "props": [("background-color", "#2E5FAC"),
                         ("color", "white"),
                         ("font-weight", "bold"),
                         ("font-size", "13px"),
                         ("text-align", "center"),
                         ("padding", "8px")]},
              {"selector": "tbody td",
               "props": [("font-size", "12px"),
                         ("padding", "6px 10px"),
                         ("text-align", "center")]},
              {"selector": "tbody td:nth-child(6)",   # Configuration column
               "props": [("text-align", "left"),
                         ("font-family", "monospace"),
                         ("font-size", "11px")]},
              {"selector": "table",
               "props": [("border-collapse", "collapse"),
                         ("width", "100%")]},
              {"selector": "tbody tr:hover",
```

```
                "props": [("filter", "brightness(0.95)")]},
            ])
            .set_caption("All 107 Grid Search Runs — Sorted by Model and Accur
)

display(styled)

# Also save as CSV
display_df.to_csv(RESULTS_DIR / "all_107_runs_display.csv", index=False)
print("Saved: all_107_runs_display.csv ✓")
```

```
Total runs: 124
```

All 107 Grid Search Runs — Sorted by Model and Accuracy

	Model	Accuracy (%)	Top-5 Accuracy (%)	Macro F1 Score	Configuration	Train Time (min)
0	EfficientNet-B0	97.39%	99.85%	0.962	img=224, ep=10, lr=0.001, opt=adam, aug=True	4.5
1	EfficientNet-B0	97.16%	99.69%	0.958	img=224, ep=10, lr=0.001, opt=adamw, aug=True	4.5
2	EfficientNet-B0	97.08%	99.92%	0.958	img=224, ep=5, lr=0.001, opt=adam, aug=True	2.3
3	EfficientNet-B0	96.93%	99.46%	0.952	img=224, ep=10, lr=0.001, opt=adamw, aug=False	4.5
4	EfficientNet-B0	96.77%	99.46%	0.953	img=224, ep=5, lr=0.001, opt=adamw, aug=False	2.3
5	EfficientNet-B0	96.77%	99.54%	0.953	img=224, ep=10, lr=0.001, opt=adam, aug=False	4.4
6	EfficientNet-B0	96.31%	99.54%	0.947	img=224, ep=5, lr=0.001, opt=adam, aug=False	2.2
7	EfficientNet-B0	96.24%	99.54%	0.944	img=224, ep=5, lr=0.001, opt=adamw, aug=True	2.3
8	EfficientNet-B0	93.86%	99.08%	0.904	img=224, ep=10, lr=0.0001, opt=adam, aug=False	4.3
9	EfficientNet-B0	93.86%	99.23%	0.899	img=224, ep=10, lr=0.0001, opt=adamw, aug=False	4.3
10	EfficientNet-B0	92.70%	98.23%	0.897	img=128, ep=10, lr=0.001, opt=adamw, aug=False	2.1
11	EfficientNet-B0	92.63%	98.69%	0.893	img=128, ep=10, lr=0.001, opt=adamw, aug=True	2.1
12	EfficientNet-B0	92.55%	98.23%	0.893	img=128, ep=10, lr=0.001, opt=adam, aug=False	2.1
13	EfficientNet-B0	92.32%	98.00%	0.890	img=128, ep=5, lr=0.001, opt=adamw, aug=True	1.1
14	EfficientNet-B0	92.32%	98.39%	0.888	img=128, ep=10, lr=0.001, opt=adam, aug=True	2.1
15	EfficientNet-B0	91.94%	98.31%	0.886	img=128, ep=5, lr=0.001, opt=adam, aug=True	1.1

	Model	Accuracy (%)	Top-5 Accuracy (%)	Macro F1 Score	Configuration	Train Time (min)
16	EfficientNet-B0	91.94%	97.85%	0.882	`img=128, ep=5, lr=0.001, opt=adam, aug=False`	1.1
17	EfficientNet-B0	91.94%	99.16%	0.875	`img=224, ep=10, lr=0.0001, opt=adam, aug=True`	4.3
18	EfficientNet-B0	91.71%	98.92%	0.867	`img=224, ep=10, lr=0.0001, opt=adamw, aug=True`	4.3
19	EfficientNet-B0	91.63%	98.23%	0.879	`img=128, ep=5, lr=0.001, opt=adamw, aug=False`	1.0
20	EfficientNet-B0	85.94%	97.39%	0.776	`img=224, ep=5, lr=0.0001, opt=adam, aug=False`	2.3
21	EfficientNet-B0	85.10%	97.16%	0.765	`img=224, ep=5, lr=0.0001, opt=adamw, aug=False`	2.3
22	EfficientNet-B0	84.33%	96.47%	0.760	`img=128, ep=10, lr=0.0001, opt=adamw, aug=False`	2.1
23	EfficientNet-B0	83.56%	96.77%	0.742	`img=224, ep=5, lr=0.0001, opt=adam, aug=True`	2.3
24	EfficientNet-B0	83.33%	95.93%	0.745	`img=128, ep=10, lr=0.0001, opt=adam, aug=False`	2.1
25	EfficientNet-B0	81.87%	96.24%	0.727	`img=224, ep=5, lr=0.0001, opt=adamw, aug=True`	2.3
26	EfficientNet-B0	81.34%	95.55%	0.672	`img=224, ep=10, lr=0.001, opt=sgd, aug=False`	4.3
27	EfficientNet-B0	80.18%	94.78%	0.685	`img=128, ep=10, lr=0.0001, opt=adamw, aug=True`	2.1
28	EfficientNet-B0	79.19%	94.55%	0.672	`img=128, ep=10, lr=0.0001, opt=adam, aug=True`	2.2
29	EfficientNet-B0	78.42%	94.01%	0.617	`img=224, ep=10, lr=0.001, opt=sgd, aug=True`	4.5
30	EfficientNet-B0	75.27%	91.32%	0.601	`img=128, ep=10, lr=0.001, opt=sgd, aug=False`	2.1
31	EfficientNet-B0	71.97%	90.71%	0.582	`img=128, ep=5, lr=0.0001, opt=adamw, aug=False`	1.1

	Model	Accuracy (%)	Top-5 Accuracy (%)	Macro F1 Score	Configuration	Train Time (min)
32	EfficientNet-B0	71.58%	90.25%	0.562	img=128, ep=5, lr=0.0001, opt=adam, aug=False	1.0
33	EfficientNet-B0	71.51%	88.25%	0.524	img=224, ep=5, lr=0.001, opt=sgd, aug=False	2.2
34	EfficientNet-B0	71.04%	89.40%	0.528	img=128, ep=10, lr=0.001, opt=sgd, aug=True	2.1
35	EfficientNet-B0	67.90%	87.40%	0.498	img=224, ep=5, lr=0.001, opt=sgd, aug=True	2.3
36	EfficientNet-B0	67.13%	88.17%	0.520	img=128, ep=5, lr=0.0001, opt=adamw, aug=True	1.1
37	EfficientNet-B0	66.90%	87.56%	0.512	img=128, ep=5, lr=0.0001, opt=adam, aug=True	1.1
38	EfficientNet-B0	62.29%	82.26%	0.397	img=128, ep=5, lr=0.001, opt=sgd, aug=False	1.0
39	EfficientNet-B0	58.53%	80.88%	0.356	img=128, ep=5, lr=0.001, opt=sgd, aug=True	1.1
40	EfficientNet-B0	36.41%	53.84%	0.126	img=224, ep=10, lr=0.0001, opt=sgd, aug=False	4.3
41	EfficientNet-B0	33.41%	51.69%	0.102	img=224, ep=10, lr=0.0001, opt=sgd, aug=True	4.3
42	EfficientNet-B0	31.18%	50.46%	0.093	img=128, ep=10, lr=0.0001, opt=sgd, aug=False	2.1
43	EfficientNet-B0	30.41%	44.24%	0.069	img=224, ep=5, lr=0.0001, opt=sgd, aug=False	2.2
44	EfficientNet-B0	30.26%	43.70%	0.070	img=224, ep=5, lr=0.0001, opt=sgd, aug=True	2.3
45	EfficientNet-B0	28.42%	44.32%	0.059	img=128, ep=10, lr=0.0001, opt=sgd, aug=True	2.1
46	EfficientNet-B0	26.73%	39.86%	0.044	img=128, ep=5, lr=0.0001, opt=sgd, aug=False	1.1
47	EfficientNet-B0	24.73%	36.71%	0.029	img=128, ep=5, lr=0.0001, opt=sgd, aug=True	1.1
48	ResNet-18	97.00%	99.77%	0.954	img=224, ep=5, lr=0.001, opt=adamw, aug=False	1.3

	Model	Accuracy (%)	Top-5 Accuracy (%)	Macro F1 Score	Configuration	Train Time (min)
49	ResNet-18	96.39%	99.46%	0.946	`img=224, ep=10, lr=0.001, opt=adam, aug=False`	2.5
50	ResNet-18	96.31%	99.31%	0.947	`img=224, ep=5, lr=0.001, opt=adam, aug=False`	1.3
51	ResNet-18	96.24%	99.46%	0.943	`img=224, ep=10, lr=0.001, opt=adamw, aug=False`	2.6
52	ResNet-18	96.16%	99.39%	0.944	`img=224, ep=10, lr=0.001, opt=adamw, aug=True`	2.7
53	ResNet-18	96.01%	99.62%	0.943	`img=224, ep=10, lr=0.001, opt=adam, aug=True`	2.7
54	ResNet-18	95.85%	99.62%	0.939	`img=224, ep=5, lr=0.001, opt=adam, aug=True`	1.4
55	ResNet-18	95.62%	99.54%	0.937	`img=224, ep=5, lr=0.001, opt=adamw, aug=True`	1.4
56	ResNet-18	94.09%	98.62%	0.916	`img=128, ep=5, lr=0.001, opt=adam, aug=False`	0.7
57	ResNet-18	94.09%	99.00%	0.903	`img=224, ep=10, lr=0.0001, opt=adam, aug=False`	2.6
58	ResNet-18	93.86%	99.23%	0.911	`img=128, ep=10, lr=0.001, opt=adam, aug=False`	1.4
59	ResNet-18	93.47%	98.39%	0.904	`img=128, ep=5, lr=0.001, opt=adamw, aug=False`	0.7
60	ResNet-18	93.39%	98.92%	0.893	`img=224, ep=10, lr=0.0001, opt=adamw, aug=False`	2.6
61	ResNet-18	93.32%	98.54%	0.900	`img=128, ep=5, lr=0.001, opt=adamw, aug=True`	0.7
62	ResNet-18	93.24%	98.62%	0.906	`img=128, ep=10, lr=0.001, opt=adamw, aug=True`	1.5
63	ResNet-18	93.16%	98.77%	0.899	`img=128, ep=10, lr=0.001, opt=adamw, aug=False`	1.4
64	ResNet-18	93.09%	98.62%	0.900	`img=128, ep=10, lr=0.001, opt=adam, aug=True`	1.4

	Model	Accuracy (%)	Top-5 Accuracy (%)	Macro F1 Score	Configuration	Train Time (min)
65	ResNet-18	92.63%	98.77%	0.885	img=224, ep=10, lr=0.0001, opt=adam, aug=True	2.8
66	ResNet-18	92.24%	98.85%	0.889	img=128, ep=5, lr=0.001, opt=adam, aug=True	0.8
67	ResNet-18	91.94%	98.69%	0.874	img=224, ep=10, lr=0.0001, opt=adamw, aug=True	2.7
68	ResNet-18	91.47%	97.85%	0.871	img=128, ep=10, lr=0.0001, opt=adamw, aug=False	1.5
69	ResNet-18	90.63%	97.62%	0.856	img=128, ep=10, lr=0.0001, opt=adam, aug=False	1.4
70	ResNet-18	90.48%	98.23%	0.842	img=224, ep=10, lr=0.001, opt=sgd, aug=False	2.6
71	ResNet-18	89.48%	98.08%	0.831	img=224, ep=10, lr=0.001, opt=sgd, aug=True	2.7
72	ResNet-18	88.71%	97.24%	0.827	img=128, ep=10, lr=0.001, opt=sgd, aug=False	1.4
73	ResNet-18	88.56%	97.31%	0.829	img=128, ep=10, lr=0.0001, opt=adamw, aug=True	1.5
74	ResNet-18	88.25%	97.62%	0.819	img=128, ep=10, lr=0.0001, opt=adam, aug=True	1.5
75	ResNet-18	86.87%	97.16%	0.776	img=224, ep=5, lr=0.0001, opt=adamw, aug=False	1.3
76	ResNet-18	86.71%	96.85%	0.794	img=128, ep=10, lr=0.001, opt=sgd, aug=True	1.5
77	ResNet-18	86.18%	96.85%	0.767	img=224, ep=5, lr=0.0001, opt=adam, aug=False	1.3
78	ResNet-18	85.79%	96.47%	0.741	img=224, ep=5, lr=0.001, opt=sgd, aug=False	1.3
79	ResNet-18	83.72%	95.93%	0.744	img=128, ep=5, lr=0.001, opt=sgd, aug=False	0.7
80	ResNet-18	83.56%	95.62%	0.735	img=128, ep=5, lr=0.0001, opt=adam, aug=False	0.7
81	ResNet-18	83.33%	95.70%	0.725	img=128, ep=5, lr=0.0001,	0.7

	Model	Accuracy (%)	Top-5 Accuracy (%)	Macro F1 Score	Configuration	Train Time (min)
					opt=adamw, aug=False	
82	ResNet-18	82.33%	95.70%	0.694	img=224, ep=5, lr=0.0001, opt=adamw, aug=True	1.4
83	ResNet-18	82.18%	94.93%	0.705	img=128, ep=5, lr=0.001, opt=sgd, aug=True	0.7
84	ResNet-18	81.80%	96.08%	0.684	img=224, ep=5, lr=0.0001, opt=adam, aug=True	1.4
85	ResNet-18	81.26%	96.01%	0.656	img=224, ep=5, lr=0.001, opt=sgd, aug=True	1.4
86	ResNet-18	78.88%	92.86%	0.654	img=128, ep=5, lr=0.0001, opt=adamw, aug=True	0.7
87	ResNet-18	78.65%	93.32%	0.647	img=128, ep=5, lr=0.0001, opt=adam, aug=True	0.7
88	ResNet-18	52.15%	68.05%	0.238	img=128, ep=10, lr=0.0001, opt=sgd, aug=False	1.4
89	ResNet-18	47.16%	64.36%	0.175	img=224, ep=10, lr=0.0001, opt=sgd, aug=False	2.6
90	ResNet-18	44.70%	61.29%	0.175	img=128, ep=10, lr=0.0001, opt=sgd, aug=True	1.5
91	ResNet-18	42.17%	60.14%	0.140	img=224, ep=10, lr=0.0001, opt=sgd, aug=True	2.7
92	ResNet-18	34.49%	51.77%	0.091	img=128, ep=5, lr=0.0001, opt=sgd, aug=False	0.7
93	ResNet-18	32.64%	46.54%	0.064	img=128, ep=5, lr=0.0001, opt=sgd, aug=True	0.7
94	ResNet-18	32.64%	48.16%	0.058	img=224, ep=5, lr=0.0001, opt=sgd, aug=False	1.3
95	ResNet-18	29.34%	43.39%	0.033	img=224, ep=5, lr=0.0001, opt=sgd, aug=True	1.3
96	SVM	67.36%	84.18%	0.531	img=64, C=10, aug=False	3.8
97	SVM	66.97%	81.72%	0.523	img=64, C=10, aug=True	9.7
98	SVM	64.52%	82.33%	0.480	img=128, C=10, aug=False	14.7

	Model	Accuracy (%)	Top-5 Accuracy (%)	Macro F1 Score	Configuration	Train Time (min)
99	SVM	63.75%	80.03%	0.460	img=128, C=10, aug=True	47.1
100	ViT-Small	98.23%	99.69%	0.974	img=224, ep=10, lr=0.001, opt=adamw, aug=False	7.5
101	ViT-Small	98.00%	99.69%	0.969	img=224, ep=10, lr=0.001, opt=adam, aug=True	7.5
102	ViT-Small	98.00%	99.92%	0.969	img=224, ep=10, lr=0.001, opt=adam, aug=False	7.5
103	ViT-Small	98.00%	99.77%	0.969	img=224, ep=10, lr=0.001, opt=adamw, aug=True	7.6
104	ViT-Small	98.00%	99.92%	0.968	img=224, ep=10, lr=0.0001, opt=adam, aug=True	7.5
105	ViT-Small	97.77%	99.85%	0.966	img=224, ep=5, lr=0.001, opt=adamw, aug=True	3.8
106	ViT-Small	97.70%	99.77%	0.966	img=224, ep=5, lr=0.001, opt=adamw, aug=False	3.8
107	ViT-Small	97.70%	99.69%	0.964	img=224, ep=10, lr=0.0001, opt=adamw, aug=True	7.5
108	ViT-Small	97.62%	99.85%	0.964	img=224, ep=5, lr=0.001, opt=adam, aug=True	3.8
109	ViT-Small	97.62%	99.46%	0.962	img=224, ep=10, lr=0.0001, opt=adamw, aug=False	7.4
110	ViT-Small	97.39%	99.62%	0.958	img=224, ep=10, lr=0.0001, opt=adam, aug=False	7.3
111	ViT-Small	97.31%	99.69%	0.958	img=224, ep=5, lr=0.001, opt=adam, aug=False	3.7
112	ViT-Small	96.70%	99.69%	0.950	img=224, ep=10, lr=0.001, opt=sgd, aug=True	7.5
113	ViT-Small	96.70%	99.46%	0.943	img=224, ep=10, lr=0.001, opt=sgd, aug=False	7.4
114	ViT-Small	96.39%	99.39%	0.932	img=224, ep=5, lr=0.0001, opt=adamw, aug=False	3.7
115	ViT-Small	96.24%	99.54%	0.934	img=224, ep=5, lr=0.0001,	3.8

	Model	Accuracy (%)	Top-5 Accuracy (%)	Macro F1 Score	Configuration	Train Time (min)
					opt=adamw, aug=True	
116	ViT-Small	95.78%	99.62%	0.925	img=224, ep=5, lr=0.0001, opt=adam, aug=True	3.7
117	ViT-Small	95.47%	99.39%	0.922	img=224, ep=5, lr=0.0001, opt=adam, aug=False	3.7
118	ViT-Small	93.24%	98.85%	0.869	img=224, ep=5, lr=0.001, opt=sgd, aug=True	3.7
119	ViT-Small	93.01%	99.23%	0.865	img=224, ep=5, lr=0.001, opt=sgd, aug=False	3.7
120	ViT-Small	58.29%	77.73%	0.323	img=224, ep=10, lr=0.0001, opt=sgd, aug=False	7.3
121	ViT-Small	52.15%	74.12%	0.248	img=224, ep=10, lr=0.0001, opt=sgd, aug=True	7.4
122	ViT-Small	38.86%	57.99%	0.125	img=224, ep=5, lr=0.0001, opt=sgd, aug=False	3.7
123	ViT-Small	35.02%	55.22%	0.095	img=224, ep=5, lr=0.0001, opt=sgd, aug=True	3.8

```
Saved: all_107_runs_display.csv ✓
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In [ ]: